Date	Hrs	Work Summary	Learning Outcome	Skl	Blk/rsk
02 Feb 2026	6	The first internship session focused on providing an introduction to mobile application development using Flutter and setting the context for the overall internship. The session began with an overview of the organization, followed by a discussion on internship objectives, duration, rules, guidelines, and expected outcomes from the interns. This helped in understanding the structure of the program and the professional expectations throughout the internship period. A high-level introduction to Flutter and the Dart programming language was provided, explaining their roles in building modern mobile applications. The mentor discussed the concept of cross-platform development and how Flutter enables developers to create applications for both Android and iOS using a single codebase. The session also included an overview of the development tools, frameworks, and workflow that will be used during the internship, giving a clear picture of how projects will be planned, developed, tested, and improved over time.	Developed a clear understanding of the fundamentals of Flutter and Dart and how they work together to build cross-platform mobile applications from a single codebase. Gained better clarity on the structure of the internship, the planned learning roadmap, and the expectations for upcoming tasks and projects. Was introduced to real-world mobile app development practices, including how applications are planned, designed, and implemented in a professional environment. Also learned the importance of UI/UX principles such as layout design, user interaction, and responsiveness, and how Flutter helps in creating scalable, efficient, and maintainable mobile applications through its widget-based architecture and modern development approach.	Flutter	No Blockers or Risks are faced since it was an Introductory sessions.
03 Feb 2026	6	Today's session focused on building a strong foundation in Flutter and the Dart programming language. The session began with the installation and configuration of the Flutter development environment, including setting up the Flutter SDK, configuring the IDE, and ensuring that the emulator/device setup was working properly. This helped in understanding the complete development workflow from setup to execution. An introduction to Dart programming was provided, covering basic syntax, variable declaration, data types, functions, and control statements such as conditional and looping structures. Practical examples were discussed to understand how these concepts are applied while writing programs. The mentor also explained the basic structure of a Flutter project, including important files and directories, and how the application execution starts from the main function. Additionally, the	Gained a strong understanding of the fundamentals of Dart programming and its significance in Flutter app development. Understood how Dart serves as the core language for building application logic, managing state, and handling user interactions within Flutter applications. Explored the basic structure of a Flutter project, including key folders and files such as main.dart, and learned how the application execution begins from the main function and runs through the widget tree. Developed a clear understanding of the widget-based architecture of Flutter and how every UI element is created using widgets. Learned the difference between Stateless and Stateful widgets and identified appropriate scenarios for using each type. Practiced implementing	Flutter	No Risks faced

		concept of widgets was introduced, emphasizing that everything in Flutter is built using widgets. The difference between various types of widgets and their role in designing the user interface was discussed, giving a clear understanding of how UI components are structured and rendered in a Flutter application.	simple UI components to understand how widgets are structured, updated, and rendered on the screen, which helped build confidence in creating basic Flutter interfaces.		
04 Feb 2026	5	The session began with a detailed overview of Flutter's origin, purpose, and its relevance in the current technology landscape. The mentor explained how Flutter was introduced to overcome the challenges of developing separate applications for Android and iOS, which traditionally required different codebases, tools, and development teams. This provided clarity on the need for cross-platform frameworks in modern app development. A comparative discussion was held between Flutter, native Android development using Android Studio with Kotlin, and React Native. Key differences were highlighted in terms of performance, development speed, UI consistency, hot reload feature, and code reusability. The mentor emphasized how Flutter's single codebase approach improves efficiency while maintaining near-native performance and a consistent user interface across platforms. The session also addressed Flutter's growing demand in the industry, explaining why startups prefer it for faster product development and cost efficiency, and how enterprises are adopting it for scalable and maintainable applications. Overall, the discussion provided a broader industry perspective and helped in understanding Flutter's practical significance in real-world mobile app development.	Gained a clear understanding of the scope and importance of Flutter in modern mobile app development and how it compares with other approaches. Understood the difference between Flutter and native Android development using Kotlin, which is limited to Android-only applications, and React Native, which relies on basic and advanced React concepts for cross-platform development. This helped in recognizing why Flutter is widely preferred for building applications for both Android and iOS using a single codebase. Developed clarity on Flutter's evolution, its growing industry adoption, and the advantages it offers in terms of performance, UI consistency, faster development, and maintainability. Also learned the basics of UI widgets and layout structures in Flutter, and how different widgets are combined and nested to design responsive and structured mobile application interfaces. This strengthened the understanding of how Flutter's widget-based architecture supports scalable and visually consistent app development.	Flutter, React , Android Studio	No Blockers
05 Feb 2026	6	The session focused on understanding the Android emulator structure. The mentor demonstrated how to configure, and run a virtual device using Android Studio and explained how the emulator integrates seamlessly with the Flutter development environment. This provided clarity on how applications are tested without needing a physical device	Learned how to set up and manage Android emulators such as Pixel 6 and Pixel 9 Pro for testing Flutter applications, including configuring virtual devices and ensuring proper integration with the Flutter environment. Gained confidence in launching and using emulators effectively for development and	Flutter, Android Studio , Git	No Blockers or Risks till now

		and how debugging can be performed efficiently. A brief introduction to APIs was also provided, explaining their importance in modern mobile applications. The mentor discussed how applications communicate with external services through APIs and how data is exchanged in JSON format. This helped in understanding how real-world apps fetch and send data such as user information, product details, or authentication responses. A sample Flutter application was executed to understand the complete development workflow—from writing code to building and running the app on the emulator. This hands-on demonstration clarified how changes in code reflect instantly using Flutter's hot reload feature. The session also introduced Git and version control concepts, including repositories, commits, and pushing code to remote platforms. It was explained that VS Code would be primarily used for writing and managing Flutter code, while Android Studio would mainly be used for emulator management and device configuration. Overall, the session provided practical exposure to the development setup and tools required for professional Flutter application development.	debugging purposes. Understood the basic concept of APIs and their significance in mobile app development, particularly how applications communicate with external servers and exchange data using formats like JSON. This provided insight into how real-world apps handle dynamic data and backend communication. Gained hands-on experience in running a sample Flutter application, which helped in understanding the complete workflow from writing code to executing it on the emulator. Additionally, developed a foundational understanding of Git version control, including repositories, commits, and pushing code, along with clarity on using VS Code as the primary development environment and Android Studio mainly for emulator management and configuration.		
06 Feb 2026	6	Today's work focused on building a strong foundation in the Dart programming language used for Flutter development. The session covered core Dart concepts such as data types, variables, and basic syntax. Key differences between final and const, as well as var and dynamic, were explained in detail to understand how Dart manages memory, ensures type safety, and handles immutability in different scenarios. This helped in identifying when each keyword should be used in real application development. Basic programming constructs were also discussed, including functions, conditional statements, and simple example programs to strengthen logical thinking and coding structure. Practical demonstrations were used to show how these concepts are applied while writing Dart code for Flutter applications. Towards the end of the session, the mentor assigned practice tasks related	Developed a clear understanding of fundamental Dart programming concepts, including data types and variable declarations used in Flutter development. Learned the key differences between final and const, as well as var and dynamic, and understood when each should be used based on immutability, type safety, and runtime behavior. Strengthened overall programming basics through discussion of simple examples and logic-building exercises. Received practice assignments to work on over the weekend to reinforce these concepts through hands-on coding, improve problem-solving ability, and build confidence in applying Dart fundamentals within Flutter applications.	Flutter, android studio	No Blockers found

		to Dart fundamentals and basic Flutter usage to be completed over the weekend. These assignments were designed to improve coding consistency, reinforce theoretical concepts through hands-on practice, and help in applying the learned topics in a more practical and structured manner.			
07 Feb 2026	6	During the weekend, I continued working on the assignments provided by my internship mentor to strengthen my understanding of Dart programming concepts. The tasks involved solving coding problems based on the seven basic data types in Dart, including **int, double, String, bool, List, Map, and Set**, along with concepts such as **var, dynamic, final, and const**. I practiced writing small programs to understand how each data type is declared, initialized, and used within Dart and Flutter applications. These exercises helped me improve my familiarity with syntax, variable handling, and basic program structure. I also focused on understanding how different data types interact within a program and how they are applied while building Flutter applications. Overall, the weekend practice helped reinforce the concepts learned during the sessions and improved my confidence in writing basic Dart code independently.	Strengthened my understanding of Dart's seven core data types through consistent hands-on coding practice and repeated problem-solving exercises. Improved confidence in writing structured Dart programs by correctly applying concepts usage in different scenarios. Practiced working with collections like lists, maps, and sets to understand how data is stored, accessed, and modified within a program. Enhanced logical thinking and debugging skills by completing the mentor-assigned tasks and testing programs with different inputs to observe output behavior. Gained better clarity on the differences between var, dynamic, final, and const and how each impacts memory usage, immutability, and runtime behavior in Dart. Overall, the practice helped in building a stronger foundation in Dart programming, improving coding consistency, and understanding how these fundamental concepts directly support Flutter app development.	Flutter	Initially faced minor confusion while differentiating between var and dynamic, while working on practice programs. These challenges were resolved through repeated practice, referring to documentation, and testing programs in the IDE.
08 Feb 2026	5	I continued working on the assignments given by my internship mentor to further improve my understanding of Dart programming concepts. I revised the seven basic data types in Dart — int, double, String, bool, List, Map, and Set — and practiced additional coding exercises to strengthen my logic and syntax clarity. I also focused on concepts like var, dynamic, final, and const to better understand when and where each should be used in real Flutter applications. I worked on writing and testing small programs that involved data storage, type conversion, and simple operations using these data types. This helped me gain more confidence in declaring variables, managing collections like lists and	Strengthened my understanding of Dart concepts by continuing the mentor-assigned assignments and practicing additional problems on data types and variables. Gained better clarity on using var, dynamic, final, and const in different scenarios within Dart programs. Improved coding confidence and logical thinking through consistent practice and debugging. Developed a deeper understanding of how these fundamental Dart concepts support Flutter app development and will be applied in building real application features.	Flutter	No Blockers

		maps, and understanding immutability in Dart. Overall, the day was spent reinforcing the previous day's learning through continuous practice and hands-on implementation of Dart fundamentals.			
09 Feb 2026	6	Today's session focused on understanding Dart operators and control flow statements used in everyday programming. The mentor explained arithmetic, comparison, and logical operators with simple examples and showed how they are applied in real coding scenarios. We also explored basic list handling and practiced using operators while working with list elements to perform operations and checks. The session then covered control flow concepts such as if–else, switch statements, and looping constructs like for, while, and do-while to manage program execution. Special attention was given to the ternary operator for writing short conditional expressions and the assert operator for debugging and validating assumptions during development. We solved a few basic programs for hands-on practice, and additional assignments were provided to further strengthen our understanding of these concepts.	Learned how to use arithmetic, comparison, and logical operators in Dart and understood how they help perform calculations, comparisons, and decision-making within programs. Gained clarity on implementing conditional statements such as if–else and switch, along with looping constructs like for, while, and do-while to effectively control the flow of application logic. Also understood how operators can be applied while working with lists and simple data handling tasks. Developed a better understanding of the ternary operator for writing concise conditions and the assert operator for debugging and validating assumptions during development. Improved overall coding confidence through hands-on practice by solving small programs during the session. Received additional assignments to continue practicing these concepts, reinforce logical thinking, and build consistency in writing structured Dart code for future Flutter application development.	Flutter	No Blockers found
10 Feb 2026	6	Today's session focused on understanding functions in Dart and their different types. The mentor explained the concept and importance of functions in structuring and reusing code. Various types of functions were discussed, including named functions, anonymous functions, arrow functions, the main function, recursive functions, and higher-order functions. The session also covered function parameters, including the difference between positional and named parameters, use of optional parameters, and default parameter values. Practical examples were demonstrated to show how these function types and parameters are used in real Flutter applications. We also received hands-on practice by solving basic programs based on functions. At	Developed a clear understanding of the different types of functions in Dart and how they are used to structure and organize code efficiently. Learned the differences between positional, named, and optional parameters, along with the use of default parameter values to make functions more flexible and readable. Gained deeper insight into advanced concepts such as recursive functions and higher-order functions, and understood their practical relevance in real programming scenarios. Improved overall coding skills through hands-on problem-solving exercises that required implementing various function types and parameter combinations.	Flutter	No Blockers found

		the end of the session, additional assignment tasks were provided to continue practicing these concepts.	Strengthened logical thinking and code structuring ability while working on practice problems. Received additional assignments to continue practicing these concepts, aiming to build stronger confidence and consistency in writing reusable and well-structured Dart code for Flutter application development.		
11 Feb 2026	6	Today's session focused on the fundamentals of Object-Oriented Programming (OOP) in Dart and how these concepts are applied in Flutter development. We learned about classes and objects, along with constructors such as default and parameterized constructors used to initialize object properties. Core OOP principles including encapsulation, inheritance, polymorphism, and abstraction were explained with simple examples to show how they help in organizing and reusing code effectively in real applications. The session also introduced the use of interfaces in Dart through the implements keyword and explained the purpose of the this and super keywords while working with classes and inheritance. Practical coding examples were demonstrated to help understand how these concepts are used in Flutter projects. We practiced writing small programs based on these topics during the session, and additional assignment tasks were provided to further strengthen our understanding through continued practice.	Gained a clear understanding of Object-Oriented Programming concepts in Dart and their importance in building structured Flutter applications. Learned how to create and use classes, objects, and constructors effectively, including both default and parameterized constructors for initializing data. Understood key OOP principles such as encapsulation, inheritance, polymorphism, and abstraction, and how these concepts support modular, reusable, and maintainable code in real-world app development. Developed knowledge of implementing interfaces using the implements keyword and learned the proper usage of this and super keywords while working with class properties and inheritance. Practiced writing small programs to apply these concepts, which helped improve code organization, logical thinking, and confidence in using OOP features within Dart and Flutter development.	Flutter	No Risks
12 Feb 2026	6	Today's session focused on understanding the difference between Stateless and Stateful widgets in Flutter and their roles in building user interfaces. The mentor explained that Stateless widgets are used when the UI does not change after being built, while Stateful widgets are required when the interface needs to update dynamically based on user interaction or data changes. A detailed walkthrough and live demonstration helped in understanding how state is created, managed, and updated within a Flutter application. We also gained hands-on experience by running a basic Hello World application	Developed a clear understanding of the difference between Stateless and Stateful widgets and identified appropriate scenarios for using each in Flutter development. Learned how state changes directly impact the user interface and gained clarity on how the setState() method triggers UI updates by rebuilding the widget tree when data changes occur. Gained practical experience in running Flutter applications on the emulator and building simple interactive applications such as a counter app with increment and	Flutter, Android Studio.	Initially faced minor confusion while understanding how setState() updates the UI and how Stateful widgets manage dynamic data. These issues were resolved through practice and guidance during the session.

		on the emulator to understand the project structure and execution flow. Additionally, we built a simple counter application with increment and decrement buttons to observe how state management works in real time. This practical implementation helped in clearly understanding how UI updates occur when setState is used and how Flutter efficiently rebuilds widgets based on state changes.	decrement functionality. This hands-on practice improved confidence in designing basic Flutter user interfaces, managing state effectively, and handling user interactions in real time.		
13 Feb 2026	6	Today's session featured an interactive quiz activity designed to review and evaluate our understanding of the topics covered so far during the internship. The quiz acted as a comprehensive recap of key concepts from Dart programming, Flutter fundamentals, widgets, Object-Oriented Programming principles, and development tools. It was conducted in an engaging and participative manner, encouraging interns to answer questions, explain their reasoning, and actively contribute to discussions. The session fostered a healthy competitive environment while reinforcing previously learned concepts and identifying areas for improvement. It helped strengthen retention of theoretical knowledge and improved confidence in recalling technical concepts quickly. At the end of the activity, points were awarded based on performance, and I secured 3rd place in the quiz, which motivated me to continue improving and performing better in upcoming sessions.	Reinforced and revised previously learned concepts from Dart and Flutter through an interactive quiz-based recap session. Strengthened understanding of key topics by actively recalling answers, participating in discussions, and applying theoretical knowledge in a time-bound setting. This helped improve confidence in quickly understanding questions and responding accurately during technical discussions. The activity also encouraged healthy competition and peer learning, which created a motivating environment to perform better. It helped identify specific areas that require further revision and practice, allowing for better focus in upcoming sessions. Overall, the session contributed to improved retention of concepts, quicker recall, and increased motivation to continue strengthening technical knowledge.	Flutter	No Blockers
14 Feb 2026	6	Today's session was dedicated to working on the assignments provided by the mentor, which involved developing two Flutter applications: a My Digital Card app and a Click Counter game app. I began by planning the UI structure and layout for the Digital Card app, where I designed a simple interface to display personal details using widgets such as Text, Container, Row, Column, and Image. This task helped in understanding widget arrangement, alignment, spacing, and styling to create a clean and structured user interface. After completing the layout design, I worked on the Click Counter game application. This involved implementing interactive buttons to increment and	Improved understanding of Flutter UI widgets and layout structures by designing and implementing the Digital Card application. Practiced arranging widgets, applying basic styling, and structuring the interface to create a clean and responsive layout. Strengthened knowledge of Stateful widgets and the use of setState() through the Click Counter game, which helped in understanding how UI updates dynamically based on user interaction. Gained hands-on experience in building simple interactive applications, testing them on the emulator, and observing real-time	Flutter , Android Studio	Faced minor issues while aligning UI elements in the Digital Card layout and managing button logic in the counter app. These were resolved through debugging, revising widget structure, and testing the app multiple times.

		decrement the count value, allowing me to practice handling user input and updating the UI dynamically using Stateful widgets. I tested both applications on the emulator to ensure proper functionality and made minor adjustments to improve layout and responsiveness. Overall, the session focused on applying previously learned Flutter concepts in a practical manner by building small but functional apps independently.	output changes. This practical implementation enhanced familiarity with the development workflow and improved confidence in applying concepts independently. Completing the assignment work also helped reinforce learning, improve coding consistency, and develop a better approach to structuring Flutter applications.		
15 Feb 2026	5	Today, I was assigned to develop a Quote of the Day application using Flutter. The task involved creating a list containing five different quotes and displaying one quote on the screen at a time. I implemented a button labeled "Next Quote" that allows users to navigate through the quotes sequentially. The logic was designed to ensure that once all quotes are displayed, the application loops back to the first quote, maintaining a continuous cycle. While working on this assignment, I used a Stateful widget to manage the current quote index and dynamically update the user interface whenever the button was pressed. This task helped reinforce concepts related to list handling, state management, button interactions, and UI rendering in Flutter. I also tested the application on the emulator to verify smooth functionality, correct state updates, and proper looping behavior. Overall, the assignment strengthened my understanding of managing dynamic content and user interaction within a Flutter application.	Enhanced understanding of using lists in Dart to store, access, and manage data efficiently within a Flutter application. Gained practical experience working with Stateful widgets and learned how to update the user interface dynamically based on user interactions such as button presses. Developed clarity on implementing button-driven logic, maintaining an index, and looping through data to create a smooth and continuous user experience. Strengthened overall confidence in building small, functional Flutter applications independently by applying concepts learned in previous sessions. Practiced testing the application on the emulator to ensure correct behavior, smooth UI updates, and proper state handling. This task also improved logical thinking, code structuring skills, and familiarity with managing dynamic content in Flutter apps.	Flutter, Android Studio	Initially faced slight confusion while implementing the looping logic to return to the first quote after reaching the last one. Also required some debugging to correctly update the quote index and refresh the UI. These issues were resolved through testing and revising the logic, which improved understanding of state handling in Flutter.
16 Feb 2026	6	Enhanced understanding of using lists in Dart to store, access, and manage data efficiently within a Flutter application. Gained practical experience working with Stateful widgets and learned how to update the user interface dynamically based on user interactions such as button presses. Developed clarity on implementing button-driven logic, maintaining an index, and looping through data to create a smooth and continuous user experience. Strengthened overall confidence in building small, functional Flutter applications independently by applying concepts learned in previous sessions.	Gained a clearer and more in-depth understanding of the differences between Stateful and Stateless widgets and their specific roles in Flutter UI development. Learned how to identify situations where a widget needs to be converted from stateless to stateful based on dynamic data changes and user interaction requirements. This helped in understanding how Flutter rebuilds the UI when state changes occur and how to structure widgets accordingly. Improved understanding of basic state management concepts,	Flutter	No Blockers

		Practiced testing the application on the emulator to ensure correct behavior, smooth UI updates, and proper state handling. This task also improved logical thinking, code structuring skills, and familiarity with managing dynamic content in Flutter apps.	including how data is stored within a widget and how UI updates are triggered. Strengthened confidence in designing interactive Flutter applications by applying these concepts through practice and small assignments. This also enhanced the ability to make better decisions while structuring widgets and managing application state effectively.		
17 Feb 2026	6	Today's session focused on implementing user input fields in Flutter applications. We learned how to use TextField widgets to take input from users and how to style them for a clear and user-friendly interface. The mentor demonstrated how to customize text fields with labels, hints, borders, and alignment to improve the overall UI. We also learned how to create a password input field where the entered text is hidden using the obscure text feature for security. The session included examples that connected previously learned concepts and calling classes within an application. We implemented a simple username and password input system, where user input was taken through text fields and handled within a Stateful widget. Hands-on practice was provided to test the functionality, style the inputs, and run the app on the emulator.	Learned how to implement and style TextField inputs in Flutter applications, including customizing properties such as decoration, labels, hints, and borders to create a clean and user-friendly interface. Understood how to configure secure password fields using obscured text to protect sensitive input and improve application security. Improved knowledge of handling user input using Stateful widgets by connecting UI components with underlying logic and managing state updates effectively. Gained practical experience in building a simple form-like interface for username and password input, which strengthened understanding of input validation basics, user interaction handling, and structured UI design in Flutter.	Flutter, Android Studio	No Blockers
18 Feb 2026	5	Today's session focused on understanding navigation in Flutter, including the use of push and pop methods to move between screens. We learned how navigation works in multi-screen applications and how data can be passed between pages. The mentor then introduced various UI feedback components such as Dialog, Alert Dialog, SnackBar, Bottom Sheet, and Toast messages, explaining their purpose and when to use each in real applications. We received hands-on practice by implementing these concepts in our existing login app with a home page. Navigation was added between login and home screens, and UI feedback elements like alert dialogs and snack bars were integrated to enhance user interaction. During the session, the mentor also gave a quick live task to add	Understood how to implement navigation in Flutter using push and pop methods to move between screens and manage the app's navigation stack effectively. Learned how to use dialog boxes, snack bars, bottom sheets, and toast messages to provide user feedback, confirmations, and alerts, helping improve the overall user experience and interaction flow within the application. Gained practical experience by integrating these features into an existing login application and testing their behavior on the emulator. This hands-on task improved understanding of when and where to use different feedback components in real scenarios. Also strengthened problem-solving and	Flutter, Android Studio	No Risks

		additional features under certain constraints, which we implemented in real time. This helped improve our speed, understanding, and confidence in applying concepts during development.	implementation skills by completing a live task within given constraints, which helped build confidence in applying concepts quickly and accurately during development.		
19 Feb 2026	6	Today's session focused on understanding ListView and GridView in Flutter. We learned the basics of creating a simple ListView and also how to build dynamic lists that display data using builder methods. The mentor explained how lists help in displaying scrollable content efficiently in mobile applications. After this, we explored GridView and its use in designing layouts such as image galleries and product displays. We continued working on the existing login app, where we integrated a GridView example on the home page. This involved adding images to the assets folder, updating the pubspec.yaml file, and displaying images in a grid layout similar to a photo gallery. A button was added on the home page to navigate and display the grid view. The mentor also introduced the physics property in Flutter to control scroll behavior and demonstrated the Stack widget, where multiple widgets can be layered on top of each other to create overlapping UI designs, further some assignment tasks were given to solve.	Gained a strong understanding of ListView and GridView and how they are used to display scrollable and grid-based data in Flutter applications. Learned how to implement dynamic lists using builder methods and how to structure UI for efficient rendering. Understood how to integrate images into an app using the assets folder and pubspec.yaml configuration, and how to display them in a GridView layout. Improved knowledge of navigation and UI structuring by extending the login app with a gallery-style grid screen. Learned how the physics property affects scrolling behavior and how the Stack widget can be used to overlay widgets and design more advanced interfaces. Through continuous demonstrations and hands-on implementation, developed better confidence in building structured and visually appealing Flutter layouts and integrating multiple widgets within a single application.	Flutter, Android Studio	Faced minor issues while configuring the assets path in pubspec.yaml and aligning grid items properly within the layout. Also required some practice to understand scroll physics behavior and widget layering using Stack. These were resolved through debugging and mentor guidance during the session.
20 Feb 2026	6	Today's session focused on understanding Inherited Widgets and Provider in Flutter and how they are used for state management across multiple widgets. The mentor explained how Inherited Widgets help share data efficiently down the widget tree without manually passing data through constructors. We also discussed the limitations of basic state management and why more structured approaches like Provider are preferred in larger applications. A detailed explanation was given on how Provider works internally, its advantages such as better scalability, cleaner code structure, and improved performance. The mentor demonstrated the practical implementation of Provider by integrating it into our existing login app. An additional feature (extra button functionality) was added to observe how shared state updates across different	Gained a clear understanding of Inherited Widgets and their role in data sharing within Flutter applications. Learned how Provider simplifies state management compared to manually managing state. Understood the advantages of Provider such as improved code maintainability, separation of concerns, and better scalability for larger apps. Developed practical knowledge of implementing Provider in a real project by extending the login app with additional functionality. Improved understanding of how state changes propagate through the widget tree and how efficient state management improves application performance and structure.	Flutter , Android Studio	No Blockers

		screens or widgets. Through this hands-on session, we understood how data flows and how UI rebuilds when state changes using Provider.			
21 Feb 2026	6	Today I worked on the assignment given by the mentor to build a Contact Manager Flutter application that demonstrates real-world multi-screen features. The task involved creating a two-screen app using navigation, ListView, dialogs, and SnackBar components. On the first screen, I implemented a contact list page displaying at least five dummy contacts using a ListView and ListTile widgets, each showing a name, phone number, and a leading person icon. A FloatingActionButton (+) was added to navigate to the second screen for adding new contacts. When a contact item is tapped, an AlertDialog was implemented to display the contact details with an OK button. On the second screen, I created an Add Contact page with two TextFields for name and phone number input and a Save button. Navigation between screens was handled using Navigator.push() and Navigator.pop(). When the Save button is pressed, a SnackBar message ("Contact Added Successfully") is displayed and the app navigates back to the contact list screen. Throughout the assignment, I focused on implementing proper navigation flow, UI layout, and interaction handling. The app was tested on the emulator to ensure that navigation, dialogs, and snack bars worked correctly.	Gained strong hands-on experience in building a multi-screen Flutter application. Learned how to use Navigator.push and pop for screen transitions and how to implement AlertDialog and SnackBar for user feedback. Improved understanding of ListView and ListTile for displaying structured data. Strengthened skills in handling user input through TextFields and managing UI interactions across screens. This assignment helped in connecting multiple Flutter concepts together into one functional mini-application, improving confidence in building real-world app features and structuring multi-screen workflows effectively. Further worked on research part and innovations add up here to improve the app interface .	Flutter, Android Studio	Faced minor issues while passing data between screens and managing navigation flow after saving a contact. Also required some debugging to correctly display dialogs and snack bars at the right time. These issues were resolved through testing and revising the navigation and UI logic.
22 Feb 2026	6	Today I worked on the second assignment, which involved building a Photo Gallery Flutter application using GridView, Stack, BottomSheet, and Navigation concepts. The objective was to create a two-screen app that simulates a gallery layout with interactive features. On the first screen (Gallery Grid Page), I implemented a GridView.count with crossAxisCount: 2 to display six colored containers representing photos (red, blue, green, orange, purple, and pink). Each grid item was designed using a Stack widget, where the background color container was overlaid with a Positioned text widget at the bottom showing labels like "Photo 1", "Photo 2", and so on. When a	Gained practical experience in using GridView.count to create structured grid-based layouts and display multiple items efficiently on screen. Improved understanding of the Stack and Positioned widgets to layer UI elements and design visually organized components like photo cards. Learned how to implement and customize a Modal BottomSheet to provide interactive options such as view, share, and delete, enhancing user experience. Strengthened knowledge of screen navigation using Navigator and passing data between screens for full-view display. Also developed better confidence in working with	Flutter, Android Studio	Faced minor issues with positioning text correctly inside the Stack and handling BottomSheet interactions properly. Also needed careful handling of navigation to ensure the correct photo data was displayed on the full-screen page. These

		grid item was tapped, I implemented a Modal BottomSheet that displayed three options: View Full, Share, and Delete, each with respective icons. Selecting "View Full" navigated to the second screen (Full View Page), where the selected colored container was displayed in full-screen mode. The AppBar included a back button and displayed the corresponding photo number as the title. Throughout the task, I focused on properly structuring the grid layout, managing navigation between screens, and implementing bottom sheet interactions. The application was tested on the emulator to ensure smooth navigation, correct UI layering with Stack, and proper functioning of the BottomSheet options.	multi-screen UI design, handling user interactions, and structuring widgets for responsive layouts. This task helped in understanding how multiple Flutter concepts work together in a real application and improved problem-solving skills while debugging layout and navigation issues. Overall, the assignment enhanced practical Flutter development skills and familiarity with building interactive, user-friendly mobile interfaces.		challenges were resolved through debugging and testing different layout adjustments.
23 Feb 2026	6	Today's session was focused on refining the previously developed applications and improving overall code structure. I revisited the Contact Manager and Photo Gallery apps to enhance UI alignment, spacing, and responsiveness. Minor improvements were made to widget structuring to ensure cleaner code and better readability. The mentor also discussed best practices for organizing Flutter projects, such as separating widgets into different files, maintaining proper naming conventions, and structuring folders efficiently. Additionally, small enhancements were implemented in the existing apps, such as improving button styling, refining layout spacing using Padding and SizedBox, and optimizing navigation flow. The session emphasized writing cleaner, maintainable code and understanding how structured design improves scalability in real-world applications.	Improved understanding of writing clean and maintainable Flutter code. Learned the importance of proper project structure and file organization. Strengthened UI design skills by refining layouts and improving responsiveness. Gained confidence in debugging and optimizing previously built applications. This session helped in understanding that app development is not only about building features but also about improving code quality, structure, and user experience for long-term scalability.	Flutter, Android Studio	Faced minor challenges in restructuring widgets without affecting functionality. Also needed to carefully test navigation flow after making modifications. These were resolved through step-by-step testing and debugging.
24 Feb 2026	7	Today's session introduced the fundamentals of networking in Flutter and how mobile applications fetch data from the internet using REST APIs. We learned how apps communicate with servers and how data is requested and received through API calls. The mentor explained the structure of a URL and API endpoint, including protocol, domain, endpoint path, and ID parameters. We also discussed common REST operations such as GET, POST, PUT, and DELETE, and their role in real-world applications. The session covered the	Gained a strong understanding of REST API concepts and how mobile apps communicate with servers over the internet. Learned how to use the http package to make API calls and how to structure URLs and endpoints properly. Developed clarity on asynchronous programming using Future, async, and await, and how these concepts ensure smooth UI performance. Improved knowledge of JSON format and parsing, including handling lists of objects and	Flutter , Android Studio	Initially found it slightly challenging to understand asynchronous flow using Future and async/await, and how data loads into the UI using FutureBuilder. Also needed some practice to correctly parse

		use of the http package in Flutter and how to set it up in a project. We explored JSON data format, how servers send responses in JSON, and how to perform JSON parsing, including parsing a list of objects. Important concepts like Future, async, and await were explained to understand asynchronous programming and how data is fetched without blocking the UI. We also learned about common HTTP status codes such as 200, 201, 400, 401, 404, and 500. For hands-on practice, we implemented API integration using free public APIs to fetch random jokes and user profiles. The fetched data was displayed in the UI using FutureBuilder, helping us understand how asynchronous data is shown in Flutter apps. Finally, we integrated API concepts into our existing login app, connecting networking logic with UI components to simulate real-world app behavior.	displaying dynamic data in the UI using FutureBuilder. Understood common HTTP status codes and their meaning in API responses. This session helped connect backend data with frontend UI, giving practical experience in integrating APIs into a Flutter application and strengthening understanding of real-world app development workflows.		JSON data and display it properly. These challenges were addressed through step-by-step examples and hands-on practice during the session.
25 Feb 2026	6	Today's session focused on implementing form validation and introducing animations in Flutter. We learned how to use a GlobalKey to manage form state and how to wrap input fields inside a Form widget. The mentor explained the difference between TextField and TextFormField, and how to add a validator function to validate user input such as empty fields, password length, and proper formatting. These validations were implemented in our existing login app to ensure correct input handling before navigation. The session then moved to Flutter animations, where we explored basic animation types such as Fade, Scale, and Slide animations. The mentor demonstrated how animations improve user experience and make the UI more interactive. We also integrated Lottie animations into the login app by adding the required dependency and updating the pubspec.yaml file. Hands-on practice was given to apply animations and validate forms within the login screen, combining both functionality and UI enhancement.	Gained practical understanding of implementing form validation using Form, GlobalKey, and TextFormField. Learned how to apply validator functions to ensure proper user input handling. Developed knowledge of basic Flutter animations such as fade, scale, and slide, and understood their importance in enhancing user experience. Improved confidence in integrating third-party packages like Lottie by updating pubspec.yaml and using animation assets effectively. Strengthened skills in combining validation logic with animated UI elements, creating a more polished and interactive login application.	Flutter, Android Studio	No Blockers
26 Feb 2026	5	Today's session focused on a comprehensive recap of all Phase 2 concepts covered so far. The mentor revised important topics including navigation operations (push and pop), ListView, GridView, TextFormField with	Revised and strengthened understanding of key Phase 2 Flutter concepts including navigation, ListView, GridView, form validation, Stack, Provider setup, and UI feedback components like	Flutter, Android Studio	The time limit of 30 seconds per question required quick decision-making, which

		validators, Stack widget, Provider and its setup methods, Inherited Widgets, Alert Dialog, SnackBar, and BottomSheet. We also revisited networking concepts such as REST API integration, JSON parsing, async/await, and animations in Flutter. The recap session helped reinforce both UI and backend-related concepts and clarified previously discussed implementations. After the revision, the mentor conducted an online quiz-based assessment through a selected website. The quiz consisted of 30 questions, with 30 seconds allotted per question, making it time-bound and challenging. The questions covered all the concepts discussed during Phase 2. The competition was intense and required quick thinking and conceptual clarity. I was able to secure 2nd place with a score of 27/30, which was both motivating and encouraging. The mentor informed us that another quiz will be conducted tomorrow and advised us to prepare thoroughly.	dialogs, snack bars, and bottom sheets. Refreshed knowledge of REST API integration, JSON parsing, async/await, and basic animations, improving clarity on how these features are applied in real applications. The quiz helped improve quick recall of concepts, time management, and confidence in answering technical questions under pressure. It also highlighted areas that need further revision and motivated preparation for the next assessment. Securing 2nd place boosted confidence since I got 3rd place last time, and encouraged active participation. The session enhanced competitive spirit, time management skills, and conceptual clarity in Flutter development.		was challenging for some scenario-based questions. However, consistent revision and practice helped in answering most questions correctly.
27 Feb 2026	5	Today's session involved a quiz-based assessment conducted by the mentor on a selected online platform. The quiz covered topics such as REST API, JSON, JSON parsing, and FutureBuilder. It consisted of 30 questions with 30 seconds per question, making it a time-bound and challenging assessment. I was able to perform well and secured 3rd place in the quiz. After a short break, the mentor informed us about a surprise MCQ quiz that would be conducted on a different platform, focusing on the topic Splash Screen in Flutter. Before starting the test, the mentor briefly revised the key highlights and important points related to splash screen implementation. After preparing quickly, we attempted the surprise quiz, and I was able to secure 3rd place again. The session concluded with the mentor assigning weekend tasks related to the concepts learned, which we are expected to complete and practice before the next session.	Strengthened understanding of REST API integration, JSON data handling, JSON parsing, and the use of FutureBuilder through a competitive quiz assessment. The timed format improved quick thinking, concept recall, and the ability to answer technical questions efficiently. The surprise quiz helped reinforce knowledge about Splash Screen implementation and encouraged quick preparation and adaptability. Participating in multiple quizzes boosted confidence, competitive spirit, and motivation to revise concepts regularly. Securing 3rd place in both quizzes reflected consistent understanding of the topics and highlighted the importance of continuous practice.	Flutter, Android Studio	No Blockers
28 Feb 2026	5	Today's session involved working on an assignment where we were asked to develop a simple Flutter application of our own choice, applying the concepts learned during the internship. The objective of the task was to design an application with proper widget structure,	This assignment helped strengthen understanding of Flutter widget structuring and UI design principles. It improved confidence in independently designing a small application from scratch and applying previously learned	Flutter, Android Studio, git	Initially required some experimentation to design the UI layout and implement the random decision

		UI layout, and interactive functionality. For this assignment, I chose to build a Decision Maker App (Spin the Choice). The app was designed to help users make quick decisions by randomly selecting an option from a predefined list. I implemented basic UI components using Flutter widgets such as Container, Text, Button, and layout widgets to structure the interface. The logic of the app was built to randomly choose an option whenever the user presses the spin or decision button. I tested the application on the emulator and refined the layout to ensure smooth functionality and clear UI presentation.	concepts such as state management, button interactions, and layout widgets. Working on this custom app also enhanced problem-solving skills by implementing random decision logic and managing UI updates dynamically. Overall, the task helped reinforce practical Flutter development skills and creativity in designing simple yet functional applications. This was great experience in finding and creating our own application of choice with proper use of all knowledges learnt till now were involved in building the app		logic effectively. Minor adjustments were needed to ensure the UI updated correctly when the decision button was pressed. These issues were resolved through testing and debugging during development.
		MARCH			
01 Mar 2026	5	Today I decided to work on a unique and advanced project idea to further improve my practical Flutter development skills. I began exploring concepts related to Firebase backend integration and studied how APIs can be used to handle dynamic data in mobile applications. This involved understanding how backend services support features such as data storage, authentication, and real-time updates in modern apps. Along with this, I started researching and planning the structure for a project titled "Digital Twin City Simulator", which is an advanced application concept that simulates a virtual representation of a city using interactive components and data-driven features. I explored how such an app could integrate APIs and backend services to handle real-time information and dynamic user interactions. This session was mainly focused on research, learning backend integration concepts, and planning the implementation approach for this project.	Gained introductory understanding of Firebase as a backend service and how it can be used in Flutter applications for storing and retrieving data. Learned how APIs interact with mobile applications to provide dynamic information and services. This exploration helped expand knowledge beyond basic UI development and introduced the importance of backend integration in building scalable and real-world mobile applications. The planning process also improved understanding of how complex applications like a Digital Twin City Simulator can be structured using Flutter and backend technologies. Even though the project currently appears complex due to ML integration and data training aspects, it will be gradually developed and refined while continuing to work on related implementations.	Flutter, Android Studio	As this project involves advanced concepts, it required additional research to understand backend integration and API handling effectively. Some complexity was encountered while planning how to structure such a large-scale application, but continued study and experimentation will help in progressing further.
02 Mar 2026	6	Today's session marked the beginning of Phase 3 concepts, where the mentor introduced Firebase and backend development concepts. Session started with a brief theoretical overview covering the history of Firebase and its role as a Backend-as-a-Service (BaaS) platform that provides ready-made backend solutions for modern applications. We learned about the Firebase services	Gained a strong conceptual understanding of Firebase as a Backend-as-a-Service platform and how it supports modern mobile and web application development. Learned about the various Firebase services that provide backend functionalities such as authentication, storage, analytics, and notifications without requiring	Flutter, Android Studio	No Blockers

		such as database management, hosting, authentication, analytics, cloud storage, notifications, and serverless backend logic, which help developers build scalable applications without managing traditional servers. The mentor also explained the difference between traditional backend architectures and Firebase's serverless approach, highlighting how Firebase simplifies development by handling infrastructure automatically. The session further covered database concepts such as the difference between SQL and NoSQL databases and how data is stored either in structured tables or in JSON-like documents. Additionally, we explored Firebase database options, including Realtime Database (RTDB) and Cloud Firestore, understanding their differences, use cases, and suitability for large-scale applications. The mentor also introduced Firestore query operations such as filtering, sorting, pagination, and compound queries, explaining how these are used to efficiently retrieve and manage data in real-world applications.	server management. Improved understanding of database structures and scaling techniques, including the differences between SQL and NoSQL databases and how data is stored and managed in Firebase. Also learned about Realtime Database and Cloud Firestore, their architecture, use cases, and how Firestore queries such as filtering, sorting, and pagination are applied to manage large datasets efficiently. This session helped build foundational knowledge required for implementing backend integration in Flutter applications.		
03 Mar 2026	6	Today's session focused on the complete Firebase setup process for integrating backend services into our Flutter application. The mentor demonstrated step-by-step how to create a Firebase project, register the Android application, and configure it properly. We learned how to copy and add the required Firebase plugin and project ID into the gradle.kts (project-level) file and how to configure the app-level gradle.kts file with necessary dependencies. The process also included downloading and placing the google-services.json file correctly inside the Android app directory. Further, we updated the pubspec.yaml file by adding dependencies for Firebase Authentication and Cloud Firestore. After completing the setup, we tested the integration using our existing login application to ensure that Firebase services were correctly connected. The mentor also briefly discussed CRUD operations (Create, Read, Update, Delete) in theory, explaining how these operations are performed in Firestore for managing application data.	Gained practical understanding of the complete Firebase setup process in a Flutter project. Learned how to configure gradle files, integrate google-services.json, and manage dependencies in pubspec.yaml. Understood how Firebase Authentication and Firestore connect with a Flutter application. Strengthened backend integration knowledge by testing Firebase within the existing login app. Also developed conceptual clarity about CRUD operations in Firestore and how data can be created, retrieved, updated, and deleted in real-world applications.	Flutter, Android Studio, Git	Initially faced minor confusion while configuring the gradle.kts files and ensuring the Firebase plugin was added at the correct project and app levels. Also required careful attention while placing the google-services.json file in the correct directory, as incorrect placement caused build errors.

04 Mar 2026	5	Today's session focused on understanding CRUD operations in Firebase Cloud Firestore. The mentor explained how data can be managed in Firestore using different operations such as Create, Read, Update, and Delete. We first learned about the create operation, where new data can be added to a collection using the .add() method. For the read operation, the mentor demonstrated how to retrieve data using .get() to fetch all stored data and .snapshots() to receive real-time updates whenever the database changes. This helped us understand how Firestore supports real-time data synchronization in applications. The session also covered update operations, where specific fields within a document can be modified without affecting the entire document. Finally, we discussed the delete operation, which allows removing particular records from the database. The mentor explained the workflow and methods used for these operations and mentioned that in the next session we will implement these CRUD operations practically in the existing login application.	Gained a clear understanding of Firestore CRUD operations and how they are used to manage application data. Learned how to add new data using .add(), retrieve data using .get() and .snapshots(), update specific fields within documents, and delete records when required. This session helped build foundational knowledge for implementing real-time database operations in Flutter applications and prepared us for the upcoming practical implementation of these concepts in the login app. It also improved understanding of how Firestore collections and documents are structured and how data flows between the Flutter application and the backend database. Overall, the session strengthened backend integration concepts and increased confidence in working with Firebase database operations within mobile applications.	Flutter, Android Studio	No Blockers
05 Mar 2026	6	Today's session focused on setting up and integrating Firebase Authentication and Cloud Firestore with our existing Flutter login application. We went through the complete configuration process and ensured that Firebase services were correctly connected to the project. The mentor demonstrated how CRUD operations can be applied using Firestore and how authentication services work alongside database storage. As part of the implementation, we added a Register feature to the existing login app where users can enter their username, email, and password. These credentials were stored in Firestore as well as Firebase Authentication, allowing users to later log in using their email and password. We tested the registration and login workflow to confirm that the data was stored and retrieved correctly. Further, the mentor assigned a few modification tasks on the login page, which I completed and demonstrated successfully. At the end of the session, the mentor introduced the idea of more complex CRUD operations that we will	Gained hands-on experience in integrating Firebase Authentication and Cloud Firestore within a Flutter application. Learned how to create a user registration system where user credentials are securely stored and used for authentication. Improved understanding of how Firestore CRUD operations interact with authentication workflows in real-world applications. This session strengthened backend integration skills and helped in understanding how user data management and login systems are implemented in mobile apps. It also prepared us for implementing more advanced database operations and data management features in upcoming sessions.	Flutter, Android Studio, Git	Initially required careful configuration of Firebase services and dependencies to ensure both authentication and Firestore worked correctly with the existing login app. Minor issues occurred while verifying the registration and login flow, but they were resolved through testing and mentor guidance during the session.

		implement in the upcoming classes to enhance backend functionality.			
06 Mar 2026	6	Today's session focused on the practical implementation of CRUD operations using Firebase Cloud Firestore. The mentor demonstrated how to manage dynamic data within the application by integrating CRUD functionality into our existing login app. For better understanding, a Student List View feature was developed where users can manage student records directly within the application. An Elevated Button was added at the bottom of the page that allows users to add new student details. On clicking the button, a form appears where users can enter information such as student name, email, age, and ID. Once the details are saved, the student record is stored in Firestore and displayed in the ListView dynamically, showing each entry one by one as they are added. Additionally, the list included edit and delete icons beside each student entry, allowing users to modify existing records or remove them when required. This approach provided a modern UI experience with clean icons and interactive design, while simultaneously handling backend database operations through Firebase.	Gained hands-on experience implementing Firestore CRUD operations in a real Flutter application. Learned how to create, display, update, and delete dynamic data using Firestore collections and documents. Improved understanding of integrating backend database functionality with interactive UI components such as ListView, buttons, and icons. This exercise also helped in understanding how modern mobile applications manage dynamic lists and user-generated data, combining clean UI design with backend data management. It strengthened practical skills in building data-driven Flutter applications using Firebase services.	Flutter, Android Studio, Git	Initially required careful handling of data updates to ensure the list refreshed correctly after adding or editing student records. Minor debugging was needed while connecting UI actions with Firestore operations, but these issues were resolved through testing and mentor guidance.
07 Mar 2026	5	Today I spent time revising the topics that were previously covered during the internship sessions to strengthen my understanding of the concepts. I reviewed key Flutter and Firebase concepts and ensured better clarity on their implementation and workflow within mobile applications. Along with revision, I also worked on self-study related to my advanced project idea – Digital Twin City Model Application. I explored various sources and collected relevant datasets that could potentially be used for this project. The focus was on understanding the structure of the datasets, analyzing different attributes, and examining how multiple datasets could be merged together. I also spent time studying the various aspects of the collected datasets, evaluating their relevance and suitability for future model training and data analysis within the application. This preparation will help in building the foundation for	Strengthened conceptual understanding by revising previously learned Flutter and Firebase concepts, reinforcing knowledge of application structure, backend integration, and database operations. The revision helped improve clarity on how different components of a Flutter application work together in real-world development. Additionally, gained initial exposure to dataset collection, exploration, and preprocessing, which are important for building data-driven applications. Learned to analyze dataset attributes, evaluate their relevance, and study how multiple datasets can be merged and structured for future machine learning model training in the Digital Twin City Model project. Overall, this activity improved research skills, data analysis	Flutter	No Blockers

		implementing the Digital Twin City model in the upcoming stages of the project.	understanding, and preparation for implementing advanced project features in upcoming stages.		
08 Mar 2026	5	Today I continued exploring Flutter development concepts, particularly focusing on the process of building APK files. I learned the difference between debug builds and release builds, where debug builds are mainly used for developer testing and debugging, while release builds (--release) are optimized for production and application distribution. This helped in understanding how Flutter applications are prepared for real user deployment. In addition to this, I spent time gaining further insights into Firebase features and how backend services can support scalable mobile applications. I also continued working on my self-learning project – Digital Twin City Model application, where I focused on improving and cleaning the datasets that I previously collected. The goal was to make the dataset more structured and optimized for future data analysis and model training. Although working with datasets and organizing them for such an advanced project is challenging, the process helped in understanding the complexity of building data-driven applications. As my Flutter development journey continues, I plan to explore more innovative features and technologies to strengthen both my mobile development and data-driven application skills.	Gained a clearer understanding of Flutter build processes, particularly the difference between debug and release builds and their role in application testing and deployment. Improved awareness of Firebase capabilities and how backend services integrate with mobile applications. Also enhanced knowledge of dataset cleaning and preparation, which is an essential step for future machine learning model development. This session contributed to continuous learning and encouraged exploration of advanced concepts in both Flutter development and data-oriented projects.	Flutter	Working with raw datasets required careful analysis and restructuring to make the data suitable for future model training. Understanding how to optimize and clean the dataset effectively required additional research and experimentation. However, continued learning and practice will help in overcoming these challenges.
09 Mar 2026	6	Today's session focused on revising the concepts discussed in the previous class and clearing doubts at the code level. The mentor provided a detailed explanation of several lines of code that previously caused confusion. We learned about the use of constructors in Dart, especially the usage of required this for initializing class variables properly. The mentor also explained how JSON data is stored in Dart using Map<String, dynamic> and why proper formatting is important to safely create object instances. In this context, the use of a factory constructor was discussed as a translator mechanism that converts JSON data into Dart objects in this student list.dart showcasing crud operations in firebase. Further, the mentor explained the importance of the	Improved understanding of Dart constructors and object initialization, particularly the use of required this for assigning values to class properties. Learned how JSON data is handled in Dart using Map<String, dynamic> and how factory constructors help convert JSON responses into Dart objects safely. Gained clarity on the role of the underscore (_) in Dart for defining private members. The debugging activity enhanced problem-solving and troubleshooting skills, especially in identifying issues related to state management and setState() usage. Overall, the session helped strengthen understanding of code structure, data handling, and	Flutter, Android Studio	Initially faced difficulty understanding how JSON data maps to Dart objects and how factory constructors simplify object creation. Additionally, identifying the exact issue causing the application to fail required careful code review, but the problem was resolved after

		underscore (_) in class names, which is commonly used to indicate private variables or classes within a Dart file. We also discussed some modern Flutter widget features and revisited concepts related to Provider for state management. During the session, we encountered an issue while running the application, which the mentor assigned as a debugging task. After carefully analyzing the problem, we identified a small mistake in the setState() definition, corrected it, and successfully ran the application.	Flutter state management practices.		correcting the setState() implementation.
10 Mar 2026	6	Today's session focused on reviewing and refining the existing Firebase Firestore CRUD implementation in the Flutter login application. The mentor explained how the Student List feature works with Firestore, where student details such as name, email, age, and grade are stored and displayed in real time using StreamBuilder. We also revised how edit, update, and delete operations function through the icons provided in each student entry. Additionally, the mentor explained the use of modal bottom sheets for adding and editing student records. Some UI concepts were also clarified, including the use of EdgeInsets.symmetric and EdgeInsets.fromLTRB for proper padding and spacing, which helps improve the layout and visual structure of the application. The session also helped in better understanding how frontend UI components interact with backend Firestore data. Overall, it strengthened confidence in working with dynamic data handling and modern Flutter UI practices.	Gained a clearer understanding of the existing Firestore CRUD implementation in the Flutter application. Reviewed how StreamBuilder is used to display real-time data from Firestore and how documents are converted into Dart objects using factory constructors. Also improved understanding of how edit, update, and delete operations interact with Firestore documents through service methods. Additionally, developed better knowledge of Flutter UI layout design, including padding and spacing using EdgeInsets.symmetric and EdgeInsets.fromLTRB. Understood how modal bottom sheets are used to display forms for adding and editing student data, improving the overall user interaction and UI structure of the application.	Flutter, Android Studio, Git	Initially required careful understanding of how Firestore streams update the UI in real time and how document IDs are used when updating or deleting records. Some attention was also needed to correctly manage widget layout spacing and form validation. These were clarified through mentor explanations and code walkthrough during the session.
11 Mar 2026	5	Today's session included a Phase 3 assessment quiz based on Firebase and backend concepts covered in recent classes. Before the quiz, we were given some time to revise the topics, after which the mentor conducted a test consisting of 40 questions with about 15 seconds per question. The questions included a mix of easy, medium, and some scenario-based problems related to Firebase setup, CRUD operations, NoSQL concepts, and database structures in RTDB and Cloud Firestore. Some questions required understanding JSON data structures and document–collection models, where collections	This quiz helped reinforce understanding of Firebase database structures, NoSQL concepts, JSON data representation, and CRUD operations in Firestore and RTDB. It also improved quick decision-making and time management skills, as each question had a strict time limit. The activity helped evaluate my current level of understanding and boosted confidence in applying backend concepts learned during Phase 3. Additionally, it strengthened my ability to analyze structured data	Flutter, Android Studio	No Blockers

		contain documents like user1, user2, with multiple fields inside them. The quiz was engaging and challenging, and I was able to score 33 out of 40 marks. One question remained unattempted due to the long scenario description. Despite this, I successfully secured 1st place in the test, marking my first top position in the ongoing assessments.	such as JSON trees and document-based database models, which are commonly used in Firebase. The quiz also highlighted the importance of accurate interpretation of scenario-based questions in technical assessments. Overall, this session helped improve both conceptual clarity and problem-solving ability related to Firebase backend operations.		
12 Mar 2026	5	Today's session focused on adding a search feature to the existing Student List in the Flutter login application. A search icon was added to the AppBar, which converts the title into a TextField when activated, allowing users to search student records by name or email. We implemented search logic using _searchQuery and _isSearching, where the _applySearch() method filters the student list based on the entered text and displays matching results dynamically. The mentor also explained how to handle cases where no matching student is found and how to display the number of filtered results. Additionally, we reviewed some UI concepts such as padding using EdgeInsets and structured lists using ListView.separated, which help maintain a clean and organized layout.	Developed a better understanding of implementing search functionality in Flutter applications using state variables and filtering logic. Learned how to dynamically filter lists using the where() method and update the UI using setState() when user input changes. Also improved knowledge of AppBar actions, TextField integration, and conditional UI rendering based on search states. The session also strengthened understanding of real-time UI updates, list filtering techniques, and improving user interaction in data-driven applications, which are essential for building modern Flutter applications.	Flutter, Android Studio, Git	No Blockers
13 Mar 2026	6	Today's session focused on adding a grade-based filtering feature to the existing Student List in the Flutter login application. We implemented ChoiceChip widgets placed horizontally to allow users to filter students based on grades such as A, B, C, D, or view all records. When a grade is selected, the application retrieves the corresponding student data from Firestore using a filtered query. The mentor explained how the where() query in Firestore is used along with orderBy() to filter students by grade and display the results dynamically through StreamBuilder. The existing search functionality was also integrated with the grade filter, allowing users to search within the filtered results. Additionally, we learned about Firebase indexing requirements when combining where and orderBy queries, and how to create indexes in Firestore for proper query execution. This session helped enhance the application by	Improved understanding of Firestore query operations, especially filtering data using where() conditions and sorting results using orderBy(). Learned how to implement UI filtering using ChoiceChip widgets and manage state changes with setState() to update the displayed data dynamically. Also gained knowledge about Firestore indexing, which is required for complex queries combining filtering and sorting. This session strengthened skills in building interactive data-driven interfaces in Flutter while efficiently retrieving filtered data from Firebase Firestore. It also helped in understanding how multiple filtering and searching mechanisms can work together within a single application feature.	Flutter, Android Studio, Git	No Blockers

		enabling dynamic filtering and better data organization in the student list.			
14 Mar 2026	6	Today I worked on an assignment to develop an Employee Management Page using Flutter integrated with Firebase Firestore. I created an Employee model class with fields such as id, name, age, department, salary, and email, along with methods like toMap() and fromFirestore() for storing and retrieving data from Firestore. An EmployeeService class was implemented to perform CRUD operations including adding, updating, fetching, and deleting employee records. The UI was designed to display employee details using Flutter cards with edit and delete options for managing records efficiently. A Floating Action Button with a form and validation was implemented to add new employees. Additionally, a search feature in the AppBar was added to filter employees by name or email, and department-based filtering using ChoiceChips was implemented to retrieve specific records from Firestore. A salary summary bar was also developed to display the total number of employees and the average salary based on the currently visible employee list, improving the usability and data insights of the application.	Gained hands-on experience in building a data-driven Flutter application integrated with Firebase Firestore. Learned how to structure model classes, implement CRUD operations, and display real-time data in the UI. Improved understanding of search and filtering techniques, form validation, and UI components such as ChoiceChips, Cards, and FloatingActionButton. This task also strengthened knowledge of Firestore queries, state management for filtering/searching, and calculating dynamic data summaries within Flutter applications. It further improved practical skills in designing interactive user interfaces while efficiently handling backend data operations.	Flutter, Android Studio, Git	No Blockers
15 Mar 2026	5	Today I worked on enhancing the Employee Management application by adding some additional improvements and bonus features suggested by the mentor. These updates were focused on refining the functionality and improving the overall user interface of the page. I also spent some time working on self-improvement tasks, where I further optimized certain parts of the application and explored ways to make the UI and functionality better. Along with this, I revised previously learned Flutter and Firebase concepts to strengthen my understanding of the implementation used in the project. I also explored and studied new features and concepts that will be covered in the upcoming class, which helped in preparing for the next session and improving overall development skills.	Improved understanding of how to enhance an existing Flutter application by adding additional features and UI refinements. Strengthened knowledge of Firestore integration, CRUD workflows, and UI component usage through revision and practical implementation. Also developed better awareness of optimizing application structure and improving usability while working on self-assigned improvements. This session helped reinforce continuous learning and preparation for upcoming advanced topics. It also improved confidence in analyzing and refining existing application features for better performance and user experience.	Flutter, Android Studio, Git	No Blockers

16 Mar 2026	6	Today's session focused on enhancing the existing Student List feature in the Flutter login application by adding advanced filtering and sorting options. We implemented a "Show More Options" feature that allows users to dynamically load additional student records using a limit-based query. The mentor explained how Firestore queries can combine filters, sorting, and limits for efficient data retrieval. We added age-based sorting functionality with ascending and descending options using an icon toggle. Additionally, a RangeSlider was implemented to filter students based on an age range, allowing users to dynamically adjust the minimum and maximum age values. The application now supports multiple query conditions, including grade filtering, age range filtering, sorting by age, and limiting the number of displayed records. A "Show More" button was also implemented to load additional students in batches, improving the usability of the list and demonstrating how pagination-like behavior can be achieved using Firestore query limits.	Gained deeper understanding of advanced Firestore query operations, including combining where, orderBy, and limit conditions in a single query. Learned how to implement dynamic filtering and sorting in Flutter using state variables and UI controls such as RangeSlider, ChoiceChips, and toggle icons. Improved knowledge of handling multiple filters simultaneously, such as grade selection, age range filtering, and sorting direction. Also understood how limit-based queries can be used to implement "Show More" functionality, which is useful for handling large datasets efficiently. This session strengthened skills in building interactive and scalable data-driven interfaces in Flutter while optimizing backend data queries using Firebase Firestore.	Flutter, Android Studio, Git	No Blockers
17 Mar 2026	5	Today's session focused on revision of previous Firebase and Firestore concepts, along with learning the difference between Spark and Blaze plans. The mentor explained how the Spark plan provides free usage limits, while the Blaze plan follows a pay-as-you-go model, including details about billing, free credits, budget alerts, and how usage is tracked. We also learned about Firebase CLI integration, which provides an alternative way to connect Firebase with a Flutter project instead of manual setup using google-services.json and gradle configuration. The mentor explained steps for upgrading to Blaze, managing billing, and setting budget alerts to avoid unexpected charges. Additionally, some video demonstrations were shown to strengthen understanding of Firestore and Flutter integration, including how .fromFirestore works similar to a translator for converting data. At the end of the session, a surprise quiz of 40 questions was conducted based on these topics. I scored 34/40, with 6 incorrect answers, and secured 1st	Gained a clear understanding of the difference between Spark and Blaze plans in Firebase, including when to use each based on project requirements. Learned how Blaze plan billing works, including pay-as-you-go pricing, budget alerts, and usage monitoring. Improved knowledge of Firebase CLI integration as an alternative to manual setup, making project configuration more efficient. Also reinforced understanding of Firestore data handling and .fromFirestore usage. This session enhanced awareness of cost management in Firebase projects, improved conceptual clarity through revision, and strengthened confidence in answering technical questions quickly during assessments. It also helped in understanding how to choose appropriate Firebase plans based on application scale and usage needs.	Flutter, Android Studio	No Blockers

		place, as bonus points were awarded for quick responses.			
18 Mar 2026	5	Today's session focused on Firebase Storage concepts and file handling in Flutter. The mentor explained the difference between Firestore and Firebase Storage, where Firestore is used as a database to store structured data like text and numbers using collection(), while Storage is used to store files such as images, videos, and PDFs using ref() as a pointer to file locations. We also learned when to use Firestore vs Storage for efficient data handling and retrieval. The session covered Firebase Storage rules, permissions setup in Android Manifest, and how files are uploaded and accessed securely. The mentor explained key operations such as putFile(), getDownloadURL(), delete(), and getMetadata(), along with the step-by-step image upload flow. Additionally, we were introduced to useful packages like image_picker, file_picker, cached_network_image, url_launcher, and methods for handling media such as displaying images efficiently with caching, opening PDFs/videos, and tracking upload progress using upload tasks.	Gained a clear understanding of the difference between Firestore and Firebase Storage and when to use each for storing structured data vs media files. Learned how to use Storage references (ref()) and perform file operations like upload, download, delete, and metadata retrieval. Improved knowledge of handling media in Flutter applications, including image selection, file picking, caching images for better performance, and opening external resources like PDFs and videos. Also understood the importance of permissions and storage rules for secure file access. This session enhanced understanding of file management in Flutter with Firebase, preparing for implementing real-world features involving media storage and retrieval.	Flutter, Android Studio	No Blockers
19 Mar 2026	5	Today I worked on the Firebase Storage implementation by upgrading the existing setup in the application. I referred to tutorials to better understand the configuration and improve the file handling features. Along with this, I did a brief revision of all topics covered so far to strengthen my overall understanding. I also focused on enhancing previous assignment projects by adding extra features and refining the UI to make them more professional and practical. This included improving structure, usability, and overall presentation of the applications. Additionally, I explored ways to make the applications more user-friendly and responsive. I also ensured that the updated features were properly tested and integrated without affecting existing functionalities.	Improved understanding of Firebase Storage setup and integration through self-learning and practical exploration. Strengthened overall knowledge by revising previously learned Flutter and Firebase concepts. Enhanced skills in refining and upgrading existing projects by adding new features and improving UI/UX. This session helped in building a more professional approach towards application development and continuous improvement. It also improved my ability to analyze existing code and identify areas for improvement. Overall, it increased confidence in independent learning and implementing advanced features in Flutter applications.	Flutter, Android Studio	No Blockers
20 Mar 2026	5	Today's session focused on the practical implementation of Firebase Storage in Flutter applications. The mentor explained in detail how file storage	Gained practical understanding of Firebase Storage integration in Flutter, including uploading and retrieving files using storage	Flutter, Android Studio, Git	Faced some difficulty while setting up Firebase

		works, including uploading and managing images, PDFs, and videos using Firebase Storage. We implemented a FileModel class to handle file details such as name, URL, type, size, and upload time, along with a readable size format. A FileService class was created to handle file upload operations using methods like putFile() and putData() for different file types. We also integrated packages such as image_picker, file_picker, cached_network, and url_launcher to select, upload, and display files efficiently within the application. Additionally, it was discussed that Firebase Storage requires a credit card setup (Blaze plan) for full usage. As an alternative, the mentor suggested using Supabase Storage in the upcoming sessions for similar functionality without billing requirements	references. Learned how to manage different file types like images, PDFs, and videos and handle their metadata effectively. Improved knowledge of using external packages such as image_picker and file_picker for file selection and cached network images for optimized display. Also understood the importance of file size handling and storage organization in real-world applications. This session also provided insight into Firebase billing requirements and alternative backend solutions like Supabase, helping in making better technology choices based on project needs.		Storage, as it requires upgrading to the Blaze plan with a credit card, which limited full implementation during practice.
21 Mar 2026	5	Today I focused on revising key Flutter and Firebase concepts to strengthen my understanding. I also attempted to set up and explore Firebase Storage with billing enabled, understanding how authentication and storage integration work together. Additionally, I started studying Supabase as an alternative backend solution, particularly its SQL-based structure, which differs from Firebase's NoSQL approach. I explored the differences in data structure, syntax, and usage patterns, and tried to understand how Supabase handles storage and database operations. I also compared how queries and data relationships are managed differently in SQL vs NoSQL systems. This helped in building a broader perspective on backend technologies used in modern applications.	Improved understanding of Flutter and Firebase concepts through revision and practical exploration. Gained initial exposure to Firebase Storage setup with billing considerations and how it integrates with application features. Also developed basic knowledge of Supabase architecture and SQL-based data handling, and how it differs from Firebase's NoSQL model. This session helped in understanding different backend approaches and choosing appropriate technologies based on project needs. It also improved awareness of data structuring techniques and query handling in relational databases. Overall, it enhanced confidence in exploring and adapting to new backend technologies independently.	Flutter, Android Studio	No Blockers
22 Mar 2026	5	Today I focused on learning Flutter concepts and working on existing projects and assignments. I spent time improving the implementation and refining the features of my ongoing work. Additionally, I identified and resolved issues related to cache accumulation, which was causing system slowdown. I cleared unnecessary cached data and optimized the environment to ensure smoother performance. I also continued enhancing my projects to make them more efficient and better structured.	Improved understanding of Flutter development through continuous practice and project work. Learned the importance of managing cache and system resources to maintain smooth performance during development. Enhanced skills in optimizing existing applications, debugging issues, and refining project structure. This session also helped in building a more efficient workflow and better system management practices while	Flutter, Android Studio	No Blockers

		Furthermore, I reviewed the project setup and dependencies to ensure everything was properly configured. I also explored ways to improve code organization and maintainability for better future development.	working on development tasks. Additionally, it improved my ability to identify performance issues and apply suitable fixes. Overall, it increased confidence in maintaining both code quality and development environment efficiency.		
23 Mar 2026	6	Today's session focused on learning Supabase as an alternative backend solution to Firebase. We started with video lectures explaining how Supabase works, including a small to-do list example to understand its functionality. The mentor explained how Supabase, being a SQL-based system, can still handle images, files, and media using URLs, unlike Firebase's NoSQL approach. Further, we were introduced to Supabase features and basic CRUD workflow, followed by a hands-on setup where we created an organization and configured a storage bucket. We connected Supabase with our Flutter project by updating pubspec.yaml and configuring Supabase URL and anon key in main.dart. Additionally, we worked on a partial implementation for uploading images, PDFs, and videos using Supabase Storage. The mentor demonstrated how files are uploaded and accessed using public URLs, and we explored handling different file types within the application.	Gained a clear understanding of Supabase architecture and how it differs from Firebase, especially in terms of SQL vs NoSQL data handling. Learned how Supabase Storage manages files using buckets and public URLs for accessing media. Improved knowledge of integrating Supabase with Flutter, including setup, configuration, and basic file upload operations. Also understood how different file types like images, PDFs, and videos are handled and retrieved efficiently. This session enhanced understanding of alternative backend solutions, strengthened skills in handling file storage using Supabase, and improved confidence in working with modern backend integrations in Flutter applications.	Flutter, Android Studio, Git	Initially required some time to understand the difference in structure and syntax between Firebase and Supabase, especially due to SQL-based handling. Also needed careful setup of Supabase credentials and bucket configuration to ensure proper connectivity. These challenges were managed through step-by-step guidance and practice.
24 Mar 2026	5	Today I focused on revising previous sessions, especially concepts related to Supabase and backend integration. The mentor conducted a surprise presentation activity where we were asked to explain Supabase concepts. It was an interactive session where everyone shared their understanding, and I also presented my points, which helped in improving both conceptual clarity and communication skills. Additionally, we worked on fixing errors in the existing Supabase-Flutter integration, ensuring proper connectivity and resolving configuration issues. This helped in stabilizing the project setup and preparing it for further implementation. I also reviewed the configuration steps to ensure proper setup for future development. The session encouraged active participation and collaborative learning among all interns.	Strengthened understanding of Supabase concepts through revision and presentation, which helped reinforce both theoretical and practical knowledge. Improved communication and presentation skills by explaining technical concepts clearly in front of others. Also gained better clarity in debugging and resolving integration issues between Supabase and Flutter. This session helped in building confidence in both technical understanding and verbal explanation of concepts, which is important for real-world development and teamwork. Additionally, it improved my ability to analyze problems quickly and present solutions effectively. Overall, it enhanced both technical confidence and interpersonal skills in a collaborative environment.	Flutter, Android Studio	No Blockers

25 Mar 2026	6	Today's session started with a revision of yesterday's Supabase Storage concepts to strengthen understanding. After that, we worked on implementing a File Manager UI design in Flutter, where we handled file operations like uploading, displaying, opening, and deleting files using Supabase Storage . The design included features such as listing uploaded files, displaying file size, using icons based on file type (image, PDF, video), and implementing floating action buttons for selecting and uploading different file formats. We also handled operations like refreshing data, opening files using URLs, and deleting files from storage. Additionally, I focused on understanding the flow of data between UI and Supabase storage. I also ensured proper testing of each feature to verify smooth functionality.	Gained better understanding of Supabase Storage integration with Flutter UI, including handling multiple file types like images, PDFs, and videos. Learned how to design a complete file manager interface with dynamic data display and user interaction. Improved knowledge of file handling operations such as upload, fetch, open, and delete, along with managing file metadata like size and type. This session also strengthened skills in UI structuring, state management, and integrating backend storage with frontend components. Additionally, it improved my ability to handle asynchronous operations effectively. Overall, it enhanced confidence in building real-world file management features in Flutter applications.	Flutter, Android Studio, Git	No Blockers
26 Mar 2026	5	Today's session focused on revising previous concepts and improving the Supabase setup. We updated the configuration by applying new storage policies for read and insert operations in the bucket to properly handle media files. This helped in resolving earlier issues related to file access and uploads. Additionally, the existing code was enhanced and refined, including better handling of file uploads and UI improvements in the File Manager application . We also worked on setting up Gmail in the emulator, where I initially faced issues due to device compatibility. After trying different devices (Pixel 6, Pixel 9a), I finally switched to a stable Pixel 6a with Android 13, which resolved the issue. Furthermore, I verified the end-to-end flow of file upload and retrieval to ensure proper working after policy updates. I also focused on testing different file types and validating permissions behavior in Supabase storage.	Gained better understanding of Supabase storage policies, especially how read and insert permissions work for secure media handling. Learned how to troubleshoot backend configuration issues and fix access-related errors effectively. Improved knowledge of enhancing existing code and integrating backend with UI smoothly. Also understood the importance of choosing the right emulator/device configuration for proper app functionality. This session strengthened skills in debugging, configuration management, and optimizing development environment setup. It also improved confidence in handling real-world issues related to backend permissions and device compatibility.	Flutter, Android Studio, Git	Faced issues with Supabase storage access due to missing or incorrect policy configuration, which initially prevented file uploads and retrieval. Also encountered problems with emulator compatibility, as Pixel 6 did not support Google Play services properly and newer versions were unstable. Additionally, setting up Gmail and testing features required switching between devices multiple times, which was time-consuming. These challenges were resolved by correcting

					storage policies and using a stable emulator configuration (Pixel 6a with Android 13).
27 Mar 2026	6	Today I focused on the completion and enhancement of the existing Supabase File Manager application. I added several modern UI improvements, making the interface more attractive and user-friendly with better layout, colors, and structured components . New features were implemented such as file preview for images, proper display of file size, and improved file cards with icons based on file type. I also added support for PDF and video uploads, along with proper handling and access through URLs. Additionally, a delete functionality with confirmation dialog was implemented to manage files efficiently. Furthermore, the application was updated to ensure smooth integration with Supabase, including handling storage paths, file metadata, and real-time UI updates after upload or deletion. I also worked on improving the overall user experience with better interaction flows and responsive design.	Gained deeper understanding of building complete file management systems using Supabase Storage and Flutter UI. Learned how to implement advanced UI components, file previews, and dynamic data display effectively. Improved knowledge of handling multiple file types (images, PDFs, videos) along with managing file metadata like size and type. Also strengthened skills in CRUD operations for storage, including upload, fetch, and delete functionalities. This session enhanced ability to design modern and interactive user interfaces, while also improving understanding of backend integration and real-time updates in Flutter applications. It also increased confidence in developing production-level features with proper user interaction and data handling.	Flutter, Android Studio, Git	No Blockers
28 Mar 2026	5	Today I focused on improving the UI widgets in the existing login page, making the design more clean, structured, and user-friendly. I worked on refining layouts, spacing, and overall component arrangement to enhance the visual appearance and usability of the application. Along with this, I spent time revising all the concepts learned this week, especially understanding how the code flows, how different widgets interact, and how data moves between UI and backend. I also analyzed the existing code to better understand its structure and improve implementation wherever needed. Additionally, I experimented with different widget combinations to achieve a more modern and responsive UI design. I also ensured that the updated changes maintained consistency across the application and did not affect existing functionalities.	Improved understanding of Flutter widget structuring and UI design principles, including layout management and component organization. Gained better clarity on application flow, state handling, and interaction between frontend and backend logic. Strengthened conceptual knowledge by revising weekly topics and applying them in real code scenarios. This session helped in building a deeper understanding of code readability, optimization, and improving existing implementations. It also enhanced confidence in modifying and improving existing applications while maintaining proper structure and functionality.	Flutter	No Blockers
29 Mar 2026	5	Today I focused on revising the complete learning journey from Dart basics to Flutter application development, covering all concepts from Phase 1,	Strengthened overall understanding of Dart, Flutter, and backend integration concepts through complete revision. Improved	Flutter, Android Studio	No Blockers

		Phase 2, and Phase 3. As we are nearing the end of the learning phase, I prepared myself for the upcoming comprehensive test by reviewing key topics and previously implemented work. Along with revision, I also spent time practicing on learning platforms to strengthen my hands-on experience and ensure better understanding of concepts. I mentally prepared for the next phase of the internship, which is the project phase, by analyzing how the learned concepts can be applied in real-world applications. Additionally, I focused on identifying weak areas and revisiting them for better clarity.	confidence in handling end-to-end application development, from basic programming to advanced features like Firebase and Supabase integration. Enhanced ability to connect theoretical concepts with practical implementation, which is essential for upcoming assessments and projects. This session also helped in building a clear roadmap for the project phase, ensuring readiness to apply learned skills effectively. Overall, it improved confidence, conceptual clarity, and readiness for both the final test and real-world project development.		
30 Mar 2026	5	Today I focused on learning how to deploy a Flutter application on the Google Play Store, including the steps involved in preparing the app for release. I explored how apps can be updated after deployment, such as version updates, bug fixes, and feature enhancements. Additionally, I researched ways to improve application quality post-deployment, including performance optimization and user experience improvements. I also explored testing and project management tools like GitHub and Jira, understanding how they are used for tracking issues, managing tasks, and maintaining app development workflows.	Gained understanding of the app deployment process on the Play Store, including release management and update cycles. Learned how continuous improvements and updates are handled after an app is published. Improved knowledge of testing strategies and tools, including how platforms like GitHub and Jira support debugging, issue tracking, and team collaboration. This session helped in understanding the complete lifecycle of a mobile application, from development to deployment and maintenance. It also enhanced awareness of industry practices for managing and improving apps after release, preparing for real-world development scenarios.	Flutter, Android Studio	No Blockers
31 Mar 2026	5	Today's session focused on revising the image picker and file upload concepts, with a deeper understanding of how they work with the Supabase backend (SQL-based). The mentor explained the practical workflow of handling media files and how they are stored and accessed through Supabase storage. We also worked on updating Supabase storage policies, where new rules were written to fix issues related to the delete functionality, which was not working earlier. After proper configuration, the delete feature was successfully implemented. Additionally, the mentor discussed that the learning phase is now complete, and we are moving towards the project phase, where we will be	Strengthened understanding of image picker integration with Supabase storage and how backend operations work in a SQL-based system. Learned how to write and manage storage policies to control access for operations like upload and delete. Improved knowledge of debugging permission-related issues and ensuring proper backend configuration. This session helped in understanding the importance of security rules in real-world applications. Overall, it marked the completion of the learning phase and prepared me for the upcoming project phase, enhancing	Flutter, Android Studio	No Blockers

		assigned projects to work on as a team and submit accordingly.	confidence in applying all learned concepts in real projects.		
		APRIL			
01 Apr 2026	5	Today's session focused on discussions regarding the upcoming project phase, including how teams will be formed and responsibilities will be assigned. We had an interactive discussion about team allocation, and currently we are waiting for the final team formation to begin the project work. The mentor also provided a Supabase-based assignment to be completed soon, which will help strengthen our practical understanding before starting the project phase. Additionally, we were informed about upcoming recorded sessions that will cover more advanced concepts and will be shared in the group for further learning. It was also emphasized that once the project phase begins, we need to maintain discipline, collaboration, and consistent contribution to successfully complete the project as a team.	Gained clarity on how the project phase will be structured, including team collaboration and task distribution. Understood the importance of working in teams, communication, and responsibility sharing in real-world development environments. Also became aware of the need for discipline and consistent effort during project development. The assignment and upcoming sessions will help in strengthening advanced concepts and preparing for practical project implementation. This session helped in mentally preparing for the transition from learning phase to project execution phase, improving readiness for real-world teamwork and development practices.	Flutter, Android Studio	No Blockers
02 Apr 2026	5	Today's session focused on project phase discussions and team formation. The mentors decided to form teams of 4 members, and further structuring and role allocation will be finalized in upcoming sessions. We were also addressed by the Chief Director of the company, who spoke about our performance, expectations, and the importance of discipline in the project phase. A brief interaction session was conducted where our communication and presentation skills were evaluated through discussions and quick responses. Additionally, various project ideas were introduced and explained by mentors, covering domains such as smart retail systems, live bus tracking, collaborative learning platforms, freelance marketplaces, alumni networking systems, and smart city issue reporting applications. Interns were also encouraged to suggest their own innovative ideas, making the session interactive and idea-driven. The final project will be selected through a	Gained clear understanding of how the project phase will be structured, including team collaboration, idea evaluation, and execution planning. Improved communication and presentation skills through interaction with mentors and leadership, which helped in expressing ideas more confidently. Also understood the importance of selecting the right project idea based on real-world impact, technical feasibility, and scalability. Exposure to multiple project domains helped in broadening thinking towards problem-solving using technology. This session enhanced readiness for the transition from learning to project implementation, strengthening both technical mindset and teamwork approach for upcoming development work.	Flutter	No Blockers

		combined discussion between mentors and interns based on feasibility, scalability, and impact.			
03 Apr 2026	6	Today's session involved working on the assigned Flutter learning task from the O'Reilly platform, where I started progressing through the course as instructed to complete it within 10 days. This helped in strengthening my understanding of Flutter concepts through structured learning. Additionally, I worked on the Shop Product Page assignment, which was successfully implemented as per the given requirements . The application included features like uploading product images, PDF price lists, and videos, displaying them with file details such as name, size, and type, and managing them through a clean UI. The assignment also included file preview, external opening using URL launcher, and delete functionality with proper UI handling, making it a real-world implementation scenario. Overall, I ensured proper integration with Supabase storage and tested all functionalities for smooth performance.	Gained practical experience in building a real-world product management page using Flutter and Supabase. Learned how to handle different file types (images, PDFs, videos) with proper upload techniques and storage handling. Improved understanding of file management workflows, including displaying file metadata, previewing content, and deleting files efficiently. Also strengthened knowledge of using packages like image_picker, file_picker, cached network images, and url_launcher in a real project scenario. This session enhanced my ability to develop complete feature-based applications, combining UI design, backend integration, and user interaction. It also improved confidence in handling assignment-based real-world problem statements independently.	Flutter, Android Studio, Git	Faced minor challenges in handling different file upload types and understanding when to use withData options. Also had some issues with managing storage paths correctly for deletion and retrieval. Additionally, ensuring proper UI updates after operations required careful state handling. These were resolved through debugging and revisiting the guidelines.
04 Apr 2026	5	Today I began the 10-day trial Flutter course assigned by the mentor to maintain discipline and track learning progress. I successfully completed Chapter 1, which mainly focused on revising the basics of Dart and Flutter. The session included understanding how Flutter uses Dart as its core programming language and how a single codebase can be used to build applications for multiple platforms like Android, iOS, Windows, macOS, and Linux. It also covered basic differences such as using Kotlin for Android and Swift for iOS in native development. Additionally, I went through setup instructions for different platforms, including Windows and macOS environments, and revised some fundamental concepts required to begin Flutter development.	Strengthened understanding of Flutter's cross-platform capabilities and how it simplifies development using a single codebase. Gained clarity on the role of Dart in Flutter and how it powers UI and logic together. Revised basic concepts which helped in refreshing the foundation for further learning. Also improved awareness of different platform setups and development environments required for building Flutter applications. This session helped in building a strong base and preparing for the upcoming chapters in the course with better clarity and confidence.	Flutter, Android Studio	No Blockers
05 Apr 2026	5	Today I completed Chapter 2 and began Chapter 3 of the Flutter course. The session mainly focused on revising and understanding the structure of a Flutter project, including the lib folder, test	Gained a clearer understanding of the Flutter project structure and file organization, which is essential for real-world development. Learned how package management works	Flutter, Android Studio	No Blockers

		folder, and metadata files, along with their purpose. I also explored how Flutter code works internally, how packages are imported, and what happens when dependencies are missing, which was demonstrated through practical video sessions. Additionally, I learned some short tricks for writing efficient Flutter and Dart code, including formatting and organizing code properly. The session also covered how widgets are structured, overridden, and centered, along with demonstrations of running a Hello World app on emulators for both Windows and macOS.	and how dependencies affect application execution. Improved knowledge of widget behavior, layout alignment, and code formatting practices. Also strengthened understanding through practical demonstrations, which helped visualize how Flutter applications run across different environments. This session enhanced overall confidence in navigating and working with Flutter projects effectively.		
06 Apr 2026	5	Today I worked on the Chapter 2 assignment, which involved building a Dice Game application using Flutter. The task focused on implementing both Stateful and Stateless widgets and understanding how they interact within an application. It was an engaging activity that helped reinforce concepts learned in Chapter 2. I also worked on integrating the assets folder to load dice images and understood how images are managed within a Flutter project. The dice values were updated dynamically using the Random function, allowing the UI to change on each interaction. Additionally, I learned how to optimize code using const where applicable for better performance. I also focused on maintaining a clean UI layout and proper widget structure during implementation. The assignment helped me understand how small applications can be built using core concepts effectively.	Strengthened understanding of Stateful vs Stateless widgets and their role in dynamic UI updates. Gained practical experience in using the Random function to create interactive applications. Improved knowledge of asset management in Flutter, including adding and displaying images correctly. Also learned the importance of code optimization using const, which helps improve performance. This assignment provided hands-on experience and helped in effectively revising and applying core Flutter concepts in a fun and practical way. It also improved my ability to build interactive UI components with proper state handling. Overall, it increased confidence in applying foundational concepts to develop small functional applications.	Flutter, Android Studio	No Blockers
07 Apr 2026	6	Today I completed Chapter 3, which focused on building a more advanced Quiz Application using Flutter. The session included setting up the app logo, implementing core logic, and designing proper data structures to manage questions and answers efficiently. I worked on creating an optimized UI, integrating Google Fonts, and handling layout issues such as overflow using proper widget structuring and wrapping techniques. The application flow was designed to smoothly navigate between questions and manage user interaction effectively. Additionally, features like adding questions and answers, shuffling questions dynamically, and tracking	Gained strong understanding of building structured applications with proper logic and data handling in Flutter. Learned how to design interactive UI with dynamic content, including managing multiple screens and user flows. Improved knowledge of handling layout challenges such as overflow and responsive design, along with using external packages like Google Fonts. Also strengthened understanding of state management for tracking user responses and results. This session enhanced my ability to build complete functional applications from scratch, combining UI, logic,	Flutter, Android Studio, Git	No Blockers

		correct and incorrect responses were implemented. The app was tested thoroughly to display final results accurately, ensuring proper functionality and user experience.	and data effectively. It also improved confidence in handling advanced Flutter concepts and real-world application scenarios.		
08 Apr 2026	6	Today I completed Chapter 4, which focused on debugging and performance analysis in Flutter applications. I learned how to run applications in debug mode, use options like start and restart, and identify errors step-by-step using the debug console. This was especially useful while working with complex apps like the quiz application involving multiple widgets. The session also introduced Flutter DevTools, including how to use profiling options in main.dart and analyze app performance through logs and visual tools in the browser. I explored features like the command palette and DevTools interface, which provide detailed insights into application behavior, UI rendering, and performance. Additionally, I learned how to track application state, inspect widgets, and make real-time changes to improve debugging efficiency. These tools made it easier to handle complex UI structures and optimize the application workflow.	Gained a strong understanding of debugging techniques in Flutter, including identifying and resolving errors using the debug console. Learned how to use Flutter DevTools for performance monitoring, logging, and UI inspection. Improved ability to analyze application behavior and optimize performance, especially in apps with multiple widgets and dynamic UI. Also understood how to use profiling and debugging tools to enhance development efficiency. This session strengthened skills in troubleshooting, performance analysis, and managing complex Flutter applications, making development more structured and efficient.	Flutter, Android Studio	No Blockers
09 Apr 2026	7	Today I worked on Chapter 5 and 6 in developing a Flutter-based Expense Tracker application to manage daily expenses efficiently. The application allows users to add, view, and delete expenses with details such as title, amount, date, and category. I implemented a structured data model using Dart classes and enums to organize expense information. The UI was designed using Material Design components with support for light and dark themes. I integrated a modal bottom sheet for adding new expenses along with form validation to ensure proper data entry. Additionally, I implemented a dynamic chart to visualize spending across different categories. The app also includes real-time UI updates using StatefulWidgets and a swipe-to-delete feature with an undo option using SnackBar. I worked on improving responsiveness by handling layout constraints and debugging UI issues to ensure smooth performance across devices.	Gained hands-on experience in building a complete Flutter application with structured data models and interactive UI. Improved understanding of state management using StatefulWidgets and real-time UI updates. Learned how to implement form validation, modal bottom sheets, and dynamic charts for better user experience. Also enhanced skills in handling themes, responsive layouts, and debugging UI issues. This project strengthened my knowledge of Flutter application architecture, user interaction design, and practical problem-solving in mobile app development. It also improved my ability to integrate multiple features into a single cohesive application. Overall, it increased confidence in developing real-world apps with proper structure and usability considerations.	Flutter, Android Studio, Git	Faced some challenges while implementing the dynamic chart and ensuring correct data mapping for categories. Also encountered minor issues with UI responsiveness and layout constraints across different screen sizes.

10 Apr 2026	9	Today I worked on Chapter 7 (Todo App) and Chapters 8, 9, 10 (Meal App) as part of my Flutter learning. In the Todo App, there was a detailed discussion on Flutter internals, including the Widget Tree, Element Tree, and Render Tree, and how they are interconnected to manage UI updates and rendering efficiently. I also learned about mutable values in memory and how Flutter handles state changes internally. The app included features like adding tasks via bottom sheet, marking tasks as complete, swipe-to-delete, and sorting todos, along with improved UI elements like priority indicators and empty state handling. In the Meal App, I followed a structured approach to build a complete application. I implemented GridView for category display, differentiation between widgets and screens, and InkWell for making widgets interactive. I worked on adding meals data, cross-screen navigation, and layout designs using Stack widgets. The app also included tab-based navigation, a side drawer for navigation, and state management using Riverpod (providers and notifier methods). Additionally, I explored explicit vs implicit animations to enhance UI transitions and user experience.	Gained deep understanding of Flutter internals, including how the Widget Tree, Element Tree, and Render Tree work together, and how state changes affect UI rendering. Learned about memory handling and mutable state, which is crucial for efficient app performance. Improved knowledge of building structured applications step-by-step, including UI components like GridView, Stack, and interactive widgets using InkWell. Enhanced understanding of navigation techniques such as tab navigation, drawer navigation, and cross-screen routing. Developed strong skills in state management using Riverpod, including providers and notifier methods for managing application data. Also learned the difference between explicit and implicit animations and how to apply them effectively in UI design. This session significantly improved my ability to build scalable, well-structured Flutter applications with proper architecture, smooth navigation, and optimized performance.	Flutter, Android Studio, Git	Faced some difficulty in understanding Flutter internals concepts like the Element Tree and Render Tree, as they required deeper conceptual clarity. Also encountered minor challenges while implementing Riverpod state management and handling navigation across multiple screens.
11 Apr 2026	10	Today I worked on Chapter 11 & 12 (Shopping List App) and Chapter 13 (Favourite Places App) as part of my Flutter learning journey. In the Shopping List App, I learned backend integration using the Firebase Realtime Database REST API with the http package. I implemented operations such as GET, POST, and DELETE for grocery items, along with proper JSON handling and response decoding. The app included features like adding items through forms with validation, loading data using FutureBuilder, swipe-to-delete using Dismissible, optimistic UI updates, and maintaining a single source of truth for state management. In the Favourite Places App, I worked on a more advanced project involving SQLite local database, Riverpod state management, image picker, GPS location services, Google Maps, and Geocoding APIs. I learned how to store places with title, image, coordinates, and address, and persist them locally using SQLite. The	Gained strong understanding of REST API integration in Flutter using Firebase Realtime Database with manual HTTP requests. Learned how to manage CRUD operations, form validation, FutureBuilder loading states, optimistic UI updates, and proper state synchronization in production-style apps. Improved knowledge of local persistence using SQLite and advanced state management using Riverpod with StateNotifierProvider. Also learned how to integrate device capabilities such as camera access, image storage, GPS location, Google Maps, and Geocoding services into Flutter applications. Developed deeper understanding of multi-screen navigation, asynchronous workflows, external package integration, and structured app architecture using models, providers, screens, and reusable	Flutter, Android Studio, Git	Faced some difficulty while understanding REST API data flow and JSON decoding, especially how response data is mapped into models. Also encountered challenges in maintaining proper state updates between FutureBuilder and local lists. In the Favourite Places App, integrating multiple features like SQLite, camera, GPS, and maps

		app also included capturing images through camera, selecting locations through GPS or interactive maps, reverse geocoding coordinates into addresses, and displaying place details with map previews and structured UI design. Additionally, both projects helped me understand multi-screen navigation, asynchronous programming, state updates, local storage, external APIs, and real-world app architecture. I also explored how to organize code into models, providers, screens, and widgets for scalable Flutter development.	widgets. This session significantly enhanced confidence in building real-world, full-featured Flutter applications with backend and device integrations. It also reinforced the importance of maintaining a single source of truth for UI state, debugging logical issues, and designing apps that are both scalable and user-friendly.		together required careful understanding of asynchronous operations and permissions handling. These challenges were resolved through practice, debugging, and revisiting concepts.
12 Apr 2026	6	Today I worked on final Chapter, building a real-time Chat Application using Flutter and Firebase as part of my internship project. The app was designed to sync messages instantly across devices using Firestore live streams, without any manual refresh. I implemented the authentication flow using Firebase Auth with login and signup on a single screen, along with form validation. During signup, user details and an auto-generated avatar URL were stored in Firestore for profile display in chats. For messaging, I used StreamBuilder to listen to the chat collection and rebuild the UI whenever new messages are added. Messages were ordered by timestamp, grouped visually for cleaner chat flow, and the text field cleared instantly after sending for better user experience. Additionally, I integrated ui-avatars.com for avatars and explored Firebase Cloud Messaging with Cloud Functions for push notifications. The app followed a clean structure where Firestore acts as the single source of truth.	Gained strong practical experience in building a real-time chat system using Flutter and Firebase. Learned how to use Firestore streams with StreamBuilder for instant UI updates and synchronization. Improved understanding of authentication workflows, including login, signup, validation, and storing user profiles in Firestore. Also learned how to improve UX through grouped messages and instant UI feedback. Enhanced knowledge of push notifications, Cloud Functions, and third-party avatar integration. This project also strengthened skills in clean architecture, asynchronous programming, and scalable app development. Overall, it increased confidence in developing production-style apps with real-time features and backend integration.	Flutter, Android Studio, Git	Since the firebase storage required credit card credential to go with proper flow I was unable to use it properly , which resulted in slow down of push notification features in the project even though all the backend and firebase tools handled properly, so working on alternative approach.
13 Apr 2026	6	Today I attended my Internship Review – Phase 1 and presented the work completed so far during the internship. I prepared a detailed PowerPoint presentation covering my learning journey, skills developed, mini projects, major applications, achievements, and future goals. The presentation also helped me revise all concepts and projects completed till date. During the review, I explained my progress in front of my college internship guide, coordinator, and other members. I presented applications built using Flutter, Firebase, and Supabase, along with quiz achievements and practical	Improved my presentation and communication skills by explaining technical projects and concepts in front of faculty members. Gained confidence in showcasing my work, achievements, and learning journey in a professional manner. The preparation process helped me revise previous topics, strengthen conceptual clarity, and reflect on my progress so far. It also improved my ability to organize technical information in a structured presentation format. Overall, this review session enhanced both my	Flutter, Android Studio	No Blockers

		learning outcomes gained throughout the internship. The session was a valuable experience that improved my confidence in presenting technical work and summarizing my internship progress professionally.	technical confidence and professional presentation abilities.		
14 Apr 2026	5	Today I focused on revising previously learned Flutter concepts and recapping the major topics covered during the internship. The session helped refresh my understanding of Flutter fundamentals, widgets, UI design, state management, Firebase integration, and project workflows. Along with the revision, there was a surprise interview-style assessment session based on overall learning progress. The discussion included areas such as technical knowledge, work ethics, deliverables & outcomes, and learning adaptability. I participated confidently and performed well in the session. This activity served as both a revision exercise and a practical evaluation of the knowledge gained so far.	Strengthened conceptual understanding by revisiting important Flutter and project development topics. Improved confidence in explaining technical concepts and responding to interview-style questions. Gained better awareness of how professional evaluations are conducted based on knowledge, adaptability, outcomes, and work ethics. Also improved communication and quick-thinking skills during the interactive session. Overall, the session enhanced both my technical confidence and readiness for future interviews or reviews.	Flutter, Android Studio	Faced minor pressure due to the unexpected nature of the interview-style session and the need to answer quickly. Some questions required recalling concepts instantly from previous lessons. These challenges were managed through revision, confidence, and clear communication.
15 Apr 2026	5	Today I learned about Play Store deployment concepts, including how Flutter applications can be prepared, published, and updated after release. I also spent time on self-learning to strengthen my understanding of Flutter, especially in areas I find challenging. Additionally, I explored Flutter open-source projects related to programs like GSoC, as I had already applied before the deadline. I reviewed project ideas and concepts to improve my practical understanding of real-world Flutter development. I also applied for GirlScript Summer of Code (GSSoC) to gain coding experience, collaborate with teams, and further improve development skills through open-source contribution opportunities.	Gained awareness of the app deployment lifecycle, including release preparation, publishing, and updates on the Play Store. Improved motivation to learn Flutter deeply by focusing on challenging concepts through self-study. Developed better understanding of how open-source programs like GSoC and GSSoC help in learning teamwork, code contribution, and industry-level workflows. Also improved research skills by exploring real project ideas and technologies. Overall, the session increased confidence in pursuing practical coding opportunities, collaborative learning, and long-term Flutter growth.	Flutter , Android Studio, Git	Faced some difficulty while understanding advanced Flutter concepts independently and identifying the right open-source projects to start with. Also needed time to explore contribution processes and expectations. These challenges were managed through research, self-learning, and continuous practice.
16 Apr 2026	5	Today I continued learning Play Store deployment concepts and explored additional features related to app publishing, updates, and improvement after release. I also spent time understanding how existing applications can be enhanced with better usability	Improved understanding of Play Store deployment workflow and post-release enhancement strategies. Learned the importance of continuously updating applications with better features and user experience improvements.	Flutter, Android Studio	No Blockers

		and new functionalities. Along with that, I worked on improving my previous Flutter projects by adding new features, refining UI elements, and testing the applications properly. I also revised earlier concepts while making these improvements to strengthen my practical knowledge. The session mainly focused on continuous enhancement, testing, and applying previously learned skills in real project scenarios.	Strengthened practical skills by revisiting old projects, adding new functionalities, and testing them carefully. Also improved confidence in debugging, refining UI, and applying concepts through hands-on practice. Overall, the session enhanced my ability to upgrade existing applications and maintain project quality through iterative improvements.		
17 Apr 2026	5	Today I continued with the learning flow and focused on understanding debugging techniques in Flutter applications. I explored how to identify errors, trace issues, and improve application performance through proper testing and debugging methods. I also learned how AI tools can be used to enhance projects, such as improving UI ideas, optimizing code structure, generating solutions, and speeding up development tasks. Along with technical learning, I prepared myself mentally for the upcoming team-based project assignments and collaborative work. The session was focused on improving development efficiency, problem-solving skills, and readiness for future project responsibilities.	Improved understanding of debugging practices and how systematic testing helps in fixing issues efficiently. Learned the value of analyzing errors carefully instead of only focusing on code writing. Gained awareness of how AI can support software development through idea generation, productivity improvement, and project enhancement. Also developed a better mindset for working in teams, adapting to tasks, and handling responsibilities. Overall, the session strengthened both my technical problem-solving ability and readiness for collaborative project work.	Flutter, Android Studio	Faced some difficulty in understanding advanced debugging scenarios and deciding the best ways to apply AI effectively in projects. Also needed time to prepare for upcoming teamwork responsibilities.
18 Apr 2026	5	Today we were assigned a task to explain the O'Reilly learning topics through demonstration and presentation. The mentor shared an APK application built for evaluation, where we had to log in and spin a wheel containing different tasks completed during the O'Reilly course. When I spun the wheel, I received the Meal App topic, specifically Chapter 9 and Chapter 10, for demonstration. Based on this, I spent time revising the related concepts, app flow, navigation, state management, and features implemented in that project. The session mainly focused on preparing for presentation, strengthening understanding of previous work, and getting ready to explain the project confidently.	Improved understanding of the Meal App project structure, including concepts from Chapters 9 and 10 such as navigation, UI flow, and feature implementation. Revising the project helped strengthen conceptual clarity and confidence. Developed better presentation readiness by preparing to explain technical work in a structured way. Also improved recall ability by revisiting previously completed chapters and practical implementations. Overall, the session enhanced both my technical revision skills and confidence for live demonstrations.	Flutter, Android Studio	Faced minor pressure due to the surprise task allocation and the need to quickly revise specific chapters for demonstration. Also needed time to recall all implemented features clearly. These challenges were managed through focused revision and preparation.
19 Apr 2026	5	Today I worked on preparing my part of the project presentation related to the Meal App. My assigned topics included Provider, Riverpod, and other important modules used in the application. I spent time revising these concepts, understanding their implementation, and organizing the content for clear	Improved understanding of state management concepts like Provider and Riverpod, including their role in handling application data efficiently. Strengthened knowledge of how these modules are applied in real Flutter projects such as the Meal App. Learned the	Flutter, Android Studio	Faced minor difficulty while revising multiple advanced topics together, especially state management and animations.

		presentation. I also focused on learning animations in Flutter, including explicit animations and implicit animations, along with how they improve user experience and UI interaction. The session helped me connect theoretical concepts with practical implementation in the Meal App project. Overall, the day was dedicated to presentation preparation, concept revision, and strengthening my understanding of advanced Flutter topics.	difference between explicit and implicit animations and how animations can enhance app usability and visual appeal. Also improved confidence in preparing technical topics for presentation and explaining concepts clearly. Overall, the session enhanced both my technical depth in Flutter concepts and readiness for project presentation.		Also needed time to structure the presentation content in a simple and clear way. These challenges were managed through focused study, revision, and practice.
20 Apr 2026	5	Today we had the presentation session for the previously assigned O'Reilly tasks. Only four members presented today, as each presentation included a detailed question-and-answer round with the mentor. The mentor asked practical and application-level questions during each presentation, making the session highly interactive and knowledge-focused. We observed presentations on projects such as the Todo App, Quiz App, Dice Game, and Expense Tracker. Each presenter explained their application flow, concepts used, and implementation details. Watching these demonstrations helped me understand different Flutter concepts and project structures more clearly. Since the discussions and Q&A for the four members took more time, my own presentation was postponed to tomorrow. I used the opportunity to further prepare and revise my assigned topics more deeply.	Gained valuable insights from different project presentations, including concepts used in the Todo App, Quiz App, Dice Game, and Expense Tracker. Observing multiple approaches improved my understanding of Flutter application development and project explanation methods. The mentor's practical and tricky questions highlighted the importance of learning topics deeply rather than only memorizing concepts. This motivated me to strengthen my preparation for my own presentation. Also improved awareness of how technical presentations are evaluated through both explanation and questioning. Overall, the session enhanced my conceptual clarity, presentation readiness, and motivation to learn in depth.	Flutter, Android Studio	Faced minor pressure knowing that the presentation involved application-level questions and detailed discussions. Also needed extra preparation time since my presentation was postponed to the next day.
21 Apr 2026	6	Today I completed my presentation session on the Meal App project. Before my turn, three other presentations were conducted on topics such as the Favourite Places App, Shopping List App, and basic Meal App widgets. This gave me additional insights into different Flutter project implementations and presentation styles. My assigned part was the main concepts from Chapter 9 and Chapter 10 of the Meal App. I explained topics such as Provider and Riverpod, their differences, and how state/data is shared and managed across the application. I also presented the animation concepts, including the difference between explicit and implicit animations and their use in Flutter UI design. The mentor appreciated my performance and said it was good and excellent. Later, we were informed about	Improved confidence in technical presentation and public speaking by explaining advanced Flutter topics in front of mentors and peers. Strengthened understanding of Provider, Riverpod, state management flow, and Flutter animations through preparation and presentation. Learned how project presentations involve both conceptual clarity and practical explanation. Also gained awareness of the upcoming collaborative project workflow, including idea selection, research, synopsis preparation, and implementation stages. This session enhanced both my technical depth and readiness for teamwork-based real-world projects.	Flutter, Android Studio , Git	No Blockers

		upcoming collaborative projects, where teams will be assigned topics through voting. Once the project is finalized, we will be given time to research, collect information, prepare a synopsis, and then begin the coding phase.			
22 Apr 2026	6	Today I focused on improving my skills and preparation for the upcoming team project phase. I spent time revising previously learned Flutter concepts, backend topics, and important modules covered during the internship. This helped me refresh my understanding before starting collaborative project work. I also continued revision on the O'Reilly platform, reviewing project-based lessons and practical concepts learned from previous chapters. The session was mainly dedicated to strengthening fundamentals and preparing myself for future team responsibilities. Additionally, I used this time to identify weak areas and revisit them for better clarity and confidence. I reviewed concepts such as state management, navigation, UI structuring, debugging, and backend integration to ensure stronger understanding. I also explored how these concepts can be applied effectively in a real team project environment. Furthermore, I focused on building a disciplined learning routine and improving consistency in self-study. This preparation helped me feel more ready to contribute meaningfully once the collaborative project tasks begin.	Strengthened understanding of previously learned Flutter, Firebase, Supabase, and project development concepts through revision. Improved readiness for the upcoming team-based project phase by refreshing technical knowledge. The O'Reilly revision helped reinforce practical application concepts and improved confidence in handling real project tasks. Also developed a better mindset for collaboration, continuous learning, and responsibility sharing in team environments. Overall, the session enhanced both my technical preparedness and confidence for future project execution. It also improved my ability to connect multiple concepts together and understand how they contribute to complete application development. Additionally, this revision increased self-confidence in handling upcoming assignments, debugging issues, and working efficiently with teammates in a professional environment.	Flutter, Android Studio	Faced minor difficulty in deciding which topics to prioritize during revision, as multiple concepts were covered throughout the internship. Also needed to balance theory recap with practical readiness for upcoming projects. At times, revisiting a wide range of concepts required extra focus and time management. These challenges were managed through structured revision and focused learning.
23 Apr 2026	5	Today's session focused on the upcoming project phase and final project selection process. Earlier, we were asked to vote for project ideas, and now four shortlisted projects were announced: Smart City, Alumni Network, Study Collaborative Platform, and Local Bus Tracker. These projects will be assigned to teams for the collaborative development phase. The mentor then discussed team formation and leadership responsibilities. We were asked who would be willing to take responsibility and serve as team leaders for the projects. I voluntarily stepped forward and expressed my interest in becoming a team leader. Along with me, three other members also volunteered for leadership roles. At present, we are	Gained better understanding of how project selection and team allocation are handled in a collaborative work environment. Learned the importance of leadership, initiative, and responsibility in managing team-based projects. By volunteering as a team leader, I developed more confidence in taking ownership and guiding project tasks. The session also improved awareness of how communication and decision-making play a key role in successful teamwork. Overall, this session strengthened my readiness for the project phase, enhanced leadership mindset, and prepared	Flutter, Android Studio	Faced minor uncertainty regarding which project would be assigned and how teams would be finalized. Also understood that leadership roles come with added responsibility and accountability. These challenges were managed with a positive

		waiting for the final allocation of teams and project assignments. The session was important in preparing us for teamwork, responsibility sharing, and the next stage of practical project execution.	me for collaborative development responsibilities.		mindset, willingness to learn, and readiness to contribute effectively once the project begins.
24 Apr 2026	5	Today the mentor officially declared the project teams and assigned project topics for the upcoming development phase. I was placed in a team with one of my female colleagues, and our assigned project is related to Smart City Issue Reporting. It was an exciting step as the project phase has now formally begun. After the team announcement, I discussed the project with my teammate regarding our upcoming work, responsibilities, and how we can complete the tasks effectively. We reviewed the project plan and understood that the work is divided into Week 1 to Week 12, where certain weeks are dedicated to development while some are reserved for corrections, review, and improvements. The session helped us begin initial planning for execution, collaboration, and task management so that we can work systematically throughout the project timeline.	Gained clear understanding of the project structure and timeline, including how tasks are distributed week-wise for smooth execution. Learned the importance of early planning and communication with teammates before starting development. Improved readiness for working in a small collaborative team, where coordination, responsibility sharing, and mutual support are essential. Also developed excitement and motivation to work on a real project related to Smart City Issue Reporting. Overall, the session strengthened my confidence in teamwork, project planning, and preparing for long-term task execution.	Flutter, Android Studio	Faced minor uncertainty in the beginning regarding project expectations and how tasks would be managed across multiple weeks. Also needed initial discussion to align with my teammate on workflow and planning.
25 Apr 2026	6	Today I discussed our Smart City Issue Reporting project with my teammate and started collecting resources, references, and ideas as an initial step for the project work. We focused on understanding the project scope and how we can begin the first phase effectively. Meanwhile, the mentors explained the task flow and evaluation process for the project phase. They informed us that there will be a detailed review meeting every weekend, where our weekly progress will be checked and evaluated based on the completed work. This made the project timeline more serious and goal-oriented. For Week 1, the assigned tasks are: Project Selection, Problem Statement, Study Existing Systems, and Finalizing Requirements. The required deliverables for this week are a Problem Statement Document, Feature List, and SRS Document. I started understanding these requirements and planning how to complete them within the deadline	Gained clear understanding of how the project workflow and weekly evaluation system will operate. Learned the importance of maintaining progress regularly and meeting deadlines in a professional project environment. Improved awareness of the initial project development stages such as defining the problem statement, studying existing systems, gathering requirements, and preparing documentation. Also understood how planning and research are essential before coding begins. Overall, this session strengthened my readiness for project execution, documentation work, and disciplined time management under deadlines.	Flutter, Android Studio	Faced some pressure regarding the weekly deadlines and review-based evaluation system, as consistent progress will be required. Also needed clarity on how to structure documents like SRS and problem statement professionally. These challenges were accepted positively, and I plan to manage them through

					proper planning, teamwork, and continuous effort.
26 Apr 2026	6	Today, as part of our Smart City Issue Reporting project, I worked on defining the problem statement, core features, MVP features, and SRS requirements for the project. This helped in building a clear foundation for the system and identifying the most important functionalities to implement first. I also spent time searching for and studying relevant research papers related to the project domain. I was able to collect five research papers, which will be useful for understanding existing solutions, methodologies, and possible improvements. Along with this, my teammate and I started reviewing those papers together and continued planning the next steps of the project. The session was focused on research, requirement analysis, and strengthening the base of our project work.	Gained better understanding of how to prepare a strong problem statement, feature planning, MVP scope, and SRS documentation for a software project. Learned the importance of defining requirements clearly before development begins. Improved research skills by identifying and collecting relevant academic papers related to the project. Studying these papers helped in understanding existing systems, technical approaches, and innovation opportunities. Also strengthened teamwork and collaboration by discussing findings with my teammate and aligning ideas for project progress. Overall, the session improved both my project planning and research-oriented thinking.	Flutter, Android Studio	No Blockers
27 Apr 2026	6	Today I studied Research Papers 1, 2, and 3 properly for the upcoming project presentation related to our Smart City Issue Reporting System. I focused on understanding the main concepts, methodologies, features, and technologies used in each paper so that we can apply useful ideas to our own project. These papers mainly covered topics such as complaint reporting systems, AI-based issue classification, smart routing, dashboards, GPS tracking, and cloud-based management. At the same time, my teammate worked on the assigned report preparation task, where the required contents and points were successfully provided by me. We coordinated well and continued progressing on both research and documentation together. Overall, the day was dedicated to research analysis, presentation preparation, and strengthening the project foundation through existing published work.	Gained better understanding of how Smart City complaint systems are developed using mobile apps, dashboards, cloud platforms, and AI technologies. Learned how features such as GPS location tracking, complaint categorization, real-time status updates, and officer management improve civic issue resolution. Improved knowledge of research-based project planning by analyzing existing solutions and identifying ideas that can be adapted to our own system. Also strengthened teamwork through coordination with my teammate on documentation work. Overall, the session enhanced my technical knowledge, research skills, and readiness for project presentation.	Flutter, Android Studio	Faced minor difficulty in understanding some advanced technical concepts from research papers and selecting the most relevant ideas for our project. Also needed time to balance both research study and report support work. These challenges were managed through reading, discussion, and focused analysis.
28 Apr 2026	6	Today I focused on studying the remaining two research papers and revising insights from all five papers collected for our Smart City Issue Reporting project. Through this analysis, I compared different approaches such as theoretical system design, Android-	Gained deeper understanding of the difference between theoretical research papers and practical implementation papers. Learned how the same problem can be solved using different platforms such as Android apps, web	Flutter, Android Studio	Faced minor difficulty in understanding some advanced AI concepts and comparing multiple papers

		based implementation, web-based solutions, and advanced AI-integrated models. This gave me a broader understanding of how similar projects are planned and developed. From the papers, I observed that some focused only on the conceptual framework, some were implemented only for Android, while others were web-based systems. Additional papers introduced ideas like multi-language support, multimodal LLM integration, and advanced AI frameworks for smarter automation and user interaction. By studying all five papers together, I now have a clearer picture of the limitations of existing systems, available technologies, and areas where our project can be improved. The session was highly useful for research clarity and innovation planning.	applications, or cross-platform tools like Flutter. Improved awareness of modern enhancements such as multi-language systems, multimodal AI, LLM integration, and advanced AI support tools that can make civic issue reporting smarter and more user-friendly. Most importantly, I developed a clear view of current system limitations and opportunities for innovation. Overall, the session enhanced my research thinking, technical vision, and confidence for upcoming project development.		with different technologies and approaches. Also needed time to identify which ideas are practical for our project scope. These challenges were managed through careful study, comparison, and focused learning.
29 Apr 2026	5	Today we had a review meeting with the CEO of the internship company regarding our project progress. I prepared thoroughly for the session, focusing on clear communication and presenting my work confidently. During the discussion, I answered questions mainly related to my major projects and presentations, explaining concepts, implementation, and learning outcomes. I received very good feedback for my responses and presentation, which boosted my confidence. The CEO also shared insights about current industry trends, emphasizing skill development, practical knowledge, and real-world application of concepts. Additionally, the session covered topics like offline deployment, local database handling, and operations, which provided a broader understanding of how applications can function beyond cloud-based systems.	Improved my communication and presentation skills by confidently explaining technical work in front of higher-level professionals. Gained clarity on how to present projects in a structured and impactful way. Learned about industry expectations and trending skills, including the importance of building practical, deployable applications. Also gained new knowledge about offline-first systems, local database management, and handling operations without relying completely on cloud services. Overall, the session enhanced my professional confidence, technical awareness, and readiness for real-world development scenarios.	Flutter, Android Studio	No Blockers
30 Apr 2026	5	Today, as a team of two, we successfully submitted the Week 1 report for our Smart City Issue Reporting project. The report included key deliverables such as the problem statement, feature list, and initial SRS work. There was also an important update from the mentors—the project timeline has been preponed, and we are now expected to complete all phases by May instead of the original schedule. This means we need to dedicate around 6–7 hours daily to keep	Gained experience in formal project submission and documentation, understanding how to present structured work professionally. The timeline change helped me realize the importance of time management, consistency, and disciplined work habits. Also improved readiness for review meetings, including how to present progress, explain decisions, and handle questions effectively. This	Flutter, Android Studio	Faced pressure due to the sudden change in project timeline, requiring more daily effort and faster progress. Also needed to balance preparation for the review

		up with the accelerated timeline. Along with submission, I also prepared myself for the upcoming weekly review meeting, where we need to present our Week 1 work, explain our approach, and provide updates on progress.	phase strengthened my mindset for working under deadlines and adapting quickly to changes. Overall, it enhanced my responsibility, planning ability, and preparedness for fast-paced project execution.		meeting along with ongoing work. These challenges are being managed through planning, consistent effort, and focusing on completing tasks step by step.
May					
01 May 2026	5	Today the scheduled meeting was postponed to tomorrow due to Labour's Day. I utilized this time effectively by focusing on revision of key concepts related to our project and previous learning. I also started preparing for the Week 2 assignment submission, reviewing requirements and planning the tasks that need to be completed. This extra time helped me organize my work better and stay aligned with the accelerated project timeline. Additionally, I revisited important modules like documentation flow, feature planning, and system design to ensure better clarity. I also outlined a rough plan for how I will approach Week 2 tasks step by step. Overall, the day was used productively for revision, preparation, and strengthening the foundation for upcoming work.	Improved conceptual clarity by revisiting important topics related to the project and previous learning. This helped reinforce understanding and confidence for upcoming tasks and reviews. Developed better time management and planning skills by utilizing unexpected free time efficiently. Also gained clarity on Week 2 requirements and how to approach them systematically. Additionally, improved my ability to self-evaluate weak areas and work on them through revision. This also helped in building consistency and discipline for the upcoming intensive project schedule. Overall, the session enhanced my preparedness, consistency, and ability to stay productive even during schedule changes.	Flutter, Android Studio	No Blockers
02 May 2026	6	Today we had the Week 1 review meeting, and I prepared thoroughly for presenting our work. Our team got the first chance to present, and this week I took the initiative to handle the presentation (as roles are rotated among teammates). I explained the Week 1 report, problem statement, features, and the literature survey of 5 research papers clearly. The presentation went smoothly, and the mentor appreciated my work. I was also asked questions regarding future challenges and project direction, which I answered confidently. Additionally, there was an interactive session where we had to give feedback	Improved my presentation and communication skills by taking initiative and explaining the project in front of the mentor and peers. Gained confidence in answering application-level and future-oriented questions related to the project. Learned how peer evaluation and feedback systems work, which helped me analyze other projects and also understand improvements needed in our own project. Observing other teams gave me new ideas and perspectives. Additionally, this session strengthened my	Flutter, Android Studio, Git	No Blockers.

		on other teams' projects using a point-based system, and similarly, others reviewed our project as well. This created a collaborative learning environment and helped us understand different project approaches.	leadership mindset, confidence in public speaking, and ability to explain research-based work clearly. Overall, it enhanced both my technical clarity and collaborative learning experience.		
03 May 2026	6	Today I worked on Week 2 tasks, focusing on UI design, system architecture, and wireframes for our Smart City Issue Reporting project. The mentor suggested using Figma, which was a new tool for me, so I spent time learning its basics through tutorials and understanding how to design UI screens effectively. I also explored how to structure UI using AI-based ideas and improvements, which helped in refining layout, user flow, and overall design thinking. Along with this, I discussed with my teammate how wireframes should be properly presented, including clarity, flow, and user interaction. Additionally, we worked on understanding and planning the architecture workflow, including how frontend, backend, and database components will interact. The day was mainly focused on design thinking, learning new tools, and progressing towards finalizing Week 2 deliverables.	Learned how to use Figma for UI/UX design, including creating layouts, components, and basic wireframes. This improved my ability to visualize application screens before development. Improved understanding of wireframing concepts and system architecture planning, including how user flow and backend processes connect in a real project. Also learned how AI tools can support better UI structuring and idea generation. Additionally, developed better design thinking and planning skills, which are essential before starting implementation. Overall, the session enhanced my UI design knowledge and project structuring ability.	Flutter, Android Studio	Faced initial difficulty while learning Figma, as it was a new tool and required time to understand features and workflow. Also needed effort to translate ideas into proper wireframe structure. Balancing UI design, architecture planning, and learning simultaneously was slightly challenging. These were managed through tutorials, discussions, and continuous practice.
04 May 2026	6	Today I continued working on Week 2 deliverables and successfully completed the system architecture design, flowchart design, and database design for our Smart City Issue Reporting project. I ensured the designs were well-structured and aligned with project requirements. I also improved my previous Figma UI designs, making them more professional, simple, and clear for better presentation and understanding. The focus was on achieving clean UI/UX and effective explanation through visuals. Additionally, I started preparing for the upcoming Friday presentation. Since it was my teammate's turn but she is unavailable, I will be handling the presentation again as a team leader. I am working on finalizing all parts of the project to ensure a smooth and confident delivery.	Gained strong understanding of system architecture, flowcharts, and database design, and how they connect to form a complete application structure. Improved ability to design systems in a more organized and professional way. Enhanced my UI/UX design skills in Figma, focusing on simplicity, clarity, and presentation quality. Also developed confidence in taking responsibility as a team leader and preparing for repeated presentations. Overall, this session improved my technical design skills, presentation readiness, and leadership ability under responsibility.	Flutter, Android Studio	Faced some pressure due to handling the presentation responsibility again as a team leader. Also needed to ensure all designs were clear, accurate, and presentation-ready. Balancing design completion and presentation preparation required careful time management. These

					challenges were managed through focused work and planning.
05 May 2026	6	Today I continued working on the project topics and focused on improving some UI features in Figma to make the design more clean, modern, and presentation-ready. I refined layouts, visual structure, and overall flow to improve clarity and professionalism in the project explanation. Along with design improvements, I spent time preparing for the upcoming project presentation, as we were informed that it will be presented in front of the Chief Director. Because of this, I focused on strengthening both the technical explanation and communication flow to ensure the presentation goes smoothly. I also revised the project concepts, architecture, and design decisions so that I can confidently explain every part during the presentation and handle questions effectively.	Improved my UI/UX design skills by refining Figma layouts and focusing on simplicity, readability, and professional presentation standards. Learned how small UI improvements can make explanations more effective and visually appealing. Enhanced my presentation preparation and communication skills by revising concepts deeply and practicing smooth explanation flow. Also developed better confidence in presenting technical work before higher authorities and professionals. Overall, the session strengthened both my design thinking and professional presentation readiness.	Flutter, Android Studio	No Blockers
06 May 2026	6	Today I focused on learning and refining the UI wireframes for both the User Dashboard and Admin Dashboard in our Smart City Issue Reporting project. I worked on understanding how to explain each screen clearly through the Figma presentation, including user flow, functionality, and design structure. I also carefully reviewed the system architecture workflow to ensure every module and connection in the project was properly understood and presented. Along with this, I analyzed the different flowcharts included in the project, as multiple process flows were involved and required detailed explanation. The session was mainly dedicated to improving presentation clarity, understanding project workflows deeply, and ensuring every design and diagram was properly aligned for explanation.	Improved understanding of dashboard UI structure and wireframe explanation techniques using Figma. Learned how to present user and admin functionalities in a clear and organized manner. Strengthened knowledge of architecture flow and flowchart analysis, including how different modules interact within the system. This also improved my ability to interpret and explain technical diagrams professionally. Overall, the session enhanced my visual presentation skills, system understanding, and confidence in explaining project workflows.	Flutter, Android Studio	No Blockers
07 May 2026	6	Today I continued revising the flowcharts and focused on preparing clear explanations for the database design of our Smart City Issue Reporting project. The database design was created using Supabase, and I worked on understanding how the data flow and table relationships function within the system. I also studied two important database structures in detail: the User	Improved understanding of database design using Supabase, including table structuring, relationships, and handling application data efficiently. Learned how user authentication data and issue-reporting data can be managed separately in a secure manner. Enhanced knowledge of RLS policies and database security	Flutter, Android Studio	Faced some difficulty while understanding the detailed behavior of RLS policies and how permissions are applied across different database

		Credentials table and the Issue Reporting table. Their fields, relationships, and purpose in the application workflow were carefully analyzed for proper explanation during presentation. Additionally, I learned about RLS (Row Level Security) policies in Supabase and how they help manage secure access control for users and data operations. The session mainly focused on database understanding, security concepts, and improving technical explanation skills.	concepts, understanding how controlled access improves application safety and user privacy. Also strengthened confidence in explaining database workflows and backend structure professionally. Overall, the session improved my backend design knowledge, database security awareness, and presentation readiness.		operations. Also needed careful analysis to explain multiple database tables clearly. These challenges were managed through revision, documentation study, and repeated analysis of the database structure.
08 May 2026	5	Today the scheduled project presentation was postponed due to the mentor's unavailability. I utilized this time productively by starting the coding phase of Week 3, mainly focusing on converting the UI designs into actual Flutter code. I carefully analyzed the project requirements and identified the technologies and setup needed for implementation. This included understanding the Supabase backend setup, authentication and database flow, along with the integration planning for Firebase Cloud Messaging (FCM) for notifications. I also started planning the basic application screens and structure, ensuring that the coding workflow aligns properly with the previously designed UI, architecture, and database models. The session mainly focused on transitioning from design to development.	Improved understanding of how to move from UI/UX design into practical coding implementation in Flutter. Learned how project planning, architecture, and backend requirements influence the actual development process. Enhanced knowledge about Supabase integration, backend preparation, and Firebase FCM setup for notification handling. Also gained better clarity on structuring the initial screens and organizing the application workflow. Overall, the session strengthened my development planning skills, backend integration understanding, and readiness for the implementation phase.	Flutter, Android Studio	No Blockers
09 May 2026	6	Today I continued working on the Week 3 implementation phase of our Smart City Issue Reporting project. I made significant progress on the UI and almost completed six major screens in Flutter, covering the core flow of the application. This early prototype was created to ensure we have a working demonstration while also managing upcoming academic commitments. On the backend side, I wrote the required SQL queries for database creation in Supabase and configured RLS (Row Level Security) policies for both database tables and storage buckets. I also set up authentication-related access rules and prepared the storage configuration for public file access where needed. In addition, the Firebase configuration file (google-services.json) was placed in the Android project to	Improved practical experience in developing multiple Flutter screens and integrating them with a structured backend. Strengthened understanding of Supabase SQL schema creation, authentication, storage buckets, and RLS security policies. Enhanced knowledge of project setup by combining Supabase and Firebase in a single Flutter application. Also developed better planning skills by creating an early prototype to balance both academic exams and internship deliverables. Additionally, this work increased my confidence in converting design and documentation into a functioning application. Overall, the session significantly improved my full-stack development skills, security	Flutter, Android Studio, Git	No Blockers

		enable Firebase services such as notifications. Since my final laboratory exams are approaching, I accelerated the prototype development so that I can later focus more effectively on refining the project and completing internship tasks with better quality.	understanding, and time management ability.		